ETL Schema Specification -- Customer Analytics v5

Meridian Capital

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This specification records the operational schema details, governance metadata, and release-aligned changes for the Customer Analytics v5 ETL domain at Meridian Capital. It is intended for internal reference by Data Engineering, Analytics, and Data Governance teams to maintain consistency of target models, maintainable lineage references, and measurable data quality outcomes. Content below is descriptive and catalog-driven, reflecting standard production practices and prior reviews. All timestamps are UTC. References align to the FY26Q3 planning cycle and continuous release cadence across environments.

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Stakeholders & Approvals

Role	Name/Team	Responsibility	Approval Status	Approval Timestamp	Reference Ticket
Executive Sponsor	K. Morgan (CAO)	Executive oversight	Approved	2026-07-07T16:12:03Z	MC-ANLT-1200
Product Owner	M. Li (CA PMO)	Roadmap & prioritizati..	Approved	2026-07-08T09:45:56Z	MC-ANLT-1222
Data Engineering Lead	A. Patel (DE-CA)	Technical ownership	Approved	2026-07-08T11:03:12Z	MC-DE-4411
Analytics Lead	S. Nguyen (BI-CA)	Consumption alignment	Approved	2026-07-09T13:27:41Z	MC-BI-3177
QA & Validation Lead	L. Romero (QA-Data)	Test coverage & sign-..	Approved	2026-07-10T18:40:25Z	MC-QA-2285
Data Governance	R. Alvarez (DG)	Stewardship & definiti..	Approved	2026-07-11T10:15:09Z	MC-DG-1034

Platform Operations	T. Chen (PlatOps)	Scheduling & platform ..	Acknowledged	2026-07-11T15:02:19Z	MC-OPS-5520
Risk & Compliance	N. Ford (Risk Analytics)	Data use compliance (..	Acknowledged	2026-07-12T08:58:06Z	MC-RISK-9007
FinOps	J. Ortiz (FinOps)	Cost observability	Acknowledged	2026-07-12T14:22:49Z	MC-FIN-2102
Change Advisory Board	CAB-Data Council	Release coordination	Approved	2026-07-13T12:30:13Z	MC-CAB-7715
Documentation Manag..	H. Kim (Tech Docs)	Editorial QA	Approved	2026-07-14T09:11:33Z	MC-DOC-6642

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v5.3.0	2026-07-12T06:50:11Z	Introduced fact_custo..	K. Zhao	MC-ANLT-1288	Moderate
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v5.2.1	2026-06-21T10:02:07Z	Extended bridge_cust..	P. Mehta	MC-BI-3221	Minor
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Operational Windows & Environments

Environment	Window (UTC)	Batch Orchestrati..	Primary Region	Secondary Region	Data Retention (A..	Notes	Last Verified
DEV	00:00-06:00	orch-ca-dev-17	us-east-2	us-west-2	30 days	Ad hoc developer r..	2026-07-16T03:02:..
QA	01:00-05:00	orch-ca-qa-21	us-east-2	us-west-2	45 days	Test data seeded ..	2026-07-16T03:05:..
UAT	02:00-06:00	orch-ca-uat-13	us-east-2	us-west-2	60 days	Business validatio..	2026-07-15T22:49:..
PROD	03:00-06:00	orch-ca-prod-31	us-east-2	us-west-2	25 months	Peak reduction alig..	2026-07-18T04:10:..
DR-STANDBY	As needed	orch-ca-dr-05	us-west-2	us-east-2	7 days	Cold standby; test ..	2026-07-01T11:35:..
SANDBOX	Variable	orch-ca-sbx-02	us-east-2	n/a	14 days	For exploratory an..	2026-07-12T12:02:..

In-Scope Data Domains (Customer Analytics v5)

Domain	Entities	Primary Sources	Steward	Notes	First Included Version
Customer Profile	dim_customer, dim_cu..	CRM-CORE, KYC-REG	DG: Customer	Unified customer ident..	v5.0.0
Customer Hierarchy	bridge_customer_addr..	CRM-CORE, ADDR-R..	DG: Customer	Many-to-many relation..	v5.0.0
Transactions	fact_customer_txn_dai..	CBS-LEDGER, CARD..	DG: Finance	Daily and line-level fac..	v5.0.0
Product	dim_product, r_ref_pr..	PROD-MDM	DG: Product	Standardized product t..	v5.0.1
Channel	dim_channel, r_ref_ch..	CHN-REG	DG: Channel	Digital and branch cha..	v5.1.3
Geography	dim_geography, r_ref_..	GEO-REG	DG: Geography	ISO-3166 aligned	v5.2.2
Campaigns	dim_marketing_camp..	MKT-AUTOMATE	DG: Marketing	Response attribution ..	v5.1.0
Engagement	fact_engagement_eve..	EVENT-HUB	DG: Channel	Session and event tax..	v5.1.2
Support	dim_support_category..	CRM-CARE	DG: Care	Case categories and i..	v5.2.0
Balances	fact_customer_balanc..	CBS-LEDGER	DG: Finance	End-of-day balances b..	v5.3.0
Reference	r_ref_currency, r_ref_r..	REF-DICT	DG: Reference	Controlled vocabularies	v5.0.2

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This section catalogs target entities in the Customer Analytics v5 schema, including grain, stewardship, and high-level lineage anchors. The intent is to standardize expectations for downstream usage across BI and analytics programs. Entities are organized by dimensional and fact orientation, with reference domains presented for controlled sets. Keys, cadence, and tickets are provided for traceability.

Target Entity Catalog -- Customer Analytics v5

Target Table	Schema	Grain	Purpose	Primary Key	SCD Type	Source Systems	Refresh Cadence ..
dim_customer	ca_dim	One row per custo..	Customer master a..	customer_key	Type 2	CRM-CORE, KYC-..	Daily DG: Custom..
dim_customer_stat..	ca_dim	One row per status..	Lifecycle status dic..	customer_status_k..	Type 1	CRM-CORE	Weekly DG: Custo..
bridge_customer_..	ca_bridge	Customer to addre..	Relationship mappi..	cust_addr_bridge_..	N/A	CRM-CORE, ADD..	Daily DG: Custom..
bridge_customer_c..	ca_bridge	Customer to chann..	Preferred/used cha..	cust_channel_brid..	N/A	CHN-REG	Daily DG: Channel..
dim_product	ca_dim	One row per produ..	Product catalog	product_key	Type 1	PROD-MDM	Weekly DG: Produ..
dim_channel	ca_dim	One row per chann..	Channel attributes	channel_key	Type 1	CHN-REG	Weekly DG: Chan..
dim_geography	ca_dim	One row per geo u..	Country/region attri..	geography_key	Type 1	GEO-REG	Monthly DG: Geog..
dim_time	ca_dim	One row per day	Calendar/date attri..	date_key	Type 1	REF-DICT	Static DG: Referen..
dim_marketing_ca..	ca_dim	One row per camp..	Campaign metadata	campaign_key	Type 1	MKT-AUTOMATE	Daily DG: Marketin..
dim_support_categ..	ca_dim	One row per categ..	Case topic taxono..	support_cat_key	Type 1	CRM-CARE	Monthly DG: Care ..
fact_customer_txn..	ca_fact	One row per custo..	Summarized trans..	cust_txn_daily_key	N/A	CBS-LEDGER, CA..	Daily DG: Finance ..
fact_customer_txn..	ca_fact	One row per trans..	Detailed line-level t..	txn_line_key	N/A	CBS-LEDGER, CA..	Intra-day DG: Fina..
fact_customer_bal..	ca_fact	One row per custo..	End-of-day balanc..	balance_snap_key	N/A	CBS-LEDGER	Daily DG: Finance ..
fact_engagement_..	ca_fact	One row per event	Engagement telem..	engagement_even..	N/A	EVENT-HUB	Hourly DG: Chann..
fact_marketing_res..	ca_fact	One row per respo..	Response attributi..	mkt_response_key	N/A	MKT-AUTOMATE	Daily DG: Marketin..
fact_support_inter..	ca_fact	One row per intera..	Support case inter..	support_interact_k..	N/A	CRM-CARE	Daily DG: Care M..
r_ref_currency	ca_ref	One row per curre..	ISO-4217 currency..	currency_key	Type 1	REF-DICT	Monthly DG: Refer..
r_ref_country	ca_ref	One row per country	ISO-3166 country l..	country_key	Type 1	REF-DICT	Quarterly DG: Geo..
r_ref_channel_type	ca_ref	One row per chann..	Channel classificat..	channel_type_key	Type 1	REF-DICT	Monthly DG: Chan..
r_ref_product_fami..	ca_ref	One row per produ..	Product family dicti..	product_family_key	Type 1	REF-DICT	Quarterly DG: Pro..
r_ref_reason_code	ca_ref	One row per reason	Reason code dictio..	reason_code_key	Type 1	REF-DICT	Monthly DG: Refer..

Column Naming & Standardization Glossary

Pattern/Term	Definition	Applies To	Example	Steward
customer_key	Surrogate key for dim_c..	Dimensions, Facts (FK)	customer_key=101233	DG: Customer
product_key	Surrogate key for dim_p..	Dimensions, Facts (FK)	product_key=55002	DG: Product
source_system_cd	Short code for origin sys..	All	source_system_cd='CR..	DG: Reference
eff_start_dt	Effective date start (incl..	SCD Type 2	eff_start_dt='2026-06-01'	DG: Governance
eff_end_dt	Effective date end (excl..	SCD Type 2	eff_end_dt='9999-12-31'	DG: Governance
curr_row_flg	Current row indicator (Y..	SCD Type 2	curr_row_flg='Y'	DG: Governance
load_ts	Ingestion timestamp	All	load_ts='2026-07-18T0..	PlatOps
update_ts	Last update timestamp	Upserted entities	update_ts='2026-07-18..	PlatOps
batch_id	Orchestration batch ide..	All	batch_id='orch-ca-prod-..	PlatOps
record_status_cd	Operational row status	All	record_status_cd='ACTI..	DG: Governance
country_cd	ISO-3166 Alpha-2 count..	Geography-linked	country_cd='US'	DG: Geography
currency_cd	ISO-4217 alpha code	Monetary facts	currency_cd='USD'	DG: Finance
amount_amt	Monetary amount colum..	Facts	txn_amount_amt=125.50	DG: Finance
quantity_qty	Quantity column suffix	Facts	item_quantity_qty=2	DG: Finance
pct_ratio	Percentage stored as d..	KPIs	response_rate_pct_rati..	DG: Marketing
hash_diff	Row-change fingerprint ..	Change detection	hash_diff='8f2a...'	DE-CA

reason_code_cd	Normalized reason code	Reference joined sets	reason_code_cd='RET'	DG: Reference

Data Quality Metrics (KPI) Summary

KPI	Target Threshold	Calculation Wind..	Owning Team	Monitoring Ticket	Current Status	Last Evaluated
Completeness -- c..	>= 99.95% populat..	30 days rolling	DE-CA	MC-DQ-5101	On Track	2026-07-17T06:30..
Referential Integrit..	<= 0.01% orphan ..	14 days rolling	QA-Data	MC-DQ-5102	On Track	2026-07-17T06:31..
Freshness -- fact_..	<= 90 minutes lag	Daily	PlatOps	MC-DQ-5103	Watch	2026-07-17T06:32..
Consistency -- curr..	100% in ISO-4217 ..	30 days rolling	DG: Finance	MC-DQ-5104	On Track	2026-07-16T23:00..
Validity -- country_..	100% conformant	Monthly	DG: Geography	MC-DQ-5105	On Track	2026-07-15T10:00..
Conformity -- chan..	>= 99.9% mapped	30 days rolling	DG: Channel	MC-DQ-5106	On Track	2026-07-16T08:45..
Accuracy -- balanc..	>= 99.9% within tol..	Weekly	DG: Finance	MC-DQ-5107	On Track	2026-07-14T12:15..
Duplicate Rate -- d..	<= 0.02%	30 days rolling	DE-CA	MC-DQ-5108	On Track	2026-07-16T05:20..
Timeliness -- camp..	100% by 06:00 UTC	Daily	DG: Marketing	MC-DQ-5109	On Track	2026-07-18T06:01..
Distribution Drift -- ..	KL divergence <= ..	14 days rolling	BI-CA	MC-DQ-5110	Watch	2026-07-17T05:55..

Page 3 -- Schema Entity Inventory and Load Characteristics

The following inventory enumerates Customer Analytics v5 entities in scope for Meridian Capital's curated analytics layer, inclusive of dimensional, fact, and bridge tables. Each entry documents grain, load pattern, lineage anchors, and stewardship metadata to support operational continuity and auditability. The review cycle for this release was completed under Change Set CA-R5.0.3 (approved by Data Governance Council on 2026-06-28). All timestamps are in UTC; SLA windows are aligned to the Analytics Refresh Window (ARW) of 00:00-06:00. Ticket references correspond to JIRA (MC-ETLCA series) and ServiceNow (SN-DATA series) queues managed by Data Engineering -- Customer Analytics (DE-CA).

Entity ID	Entity Name	Table Name	Layer	Grain	Description	Primary Key(s)	Surrogate Key So..
CA-DIM-001	Customer Dimensi..	dim_customer	Curated	Customer (SCD2)	Core customer pro..	customer_id	customer_sk CRM..
CA-DIM-002	Customer Status	dim_customer_stat..	Curated	Customer Status	Status code taxon..	status_code	status_sk CRM-Gl..
CA-DIM-003	Household Dimens..	dim_household	Curated	Household (SCD2)	Household groupin..	household_id	household_sk CR..
CA-DIM-004	Address Dimension	dim_address	Curated	Address (SCD2)	Standardized post..	address_id	address_sk Addre..
CA-DIM-005	Channel Dimension	dim_channel	Curated	Channel	Interaction channel..	channel_code	channel_sk MktgH..
CA-DIM-006	Segment Dimension	dim_segment	Curated	Segment	Analytical segment..	segment_code	segment_sk SegE..
CA-DIM-007	Product Reference	dim_product_refer..	Curated	Product	Product taxonomy ..	product_code	product_sk ProdC..
CA-DIM-008	Marketing Permissi..	dim_marketing_pe..	Curated	Customer (SCD2)	Consent and permi..	permission_id	permission_sk Con..
CA-DIM-009	Employment	dim_employment	Curated	Customer Employ..	Employment attrib..	employment_id	employment_sk O..
CA-DIM-010	KYC Flags	dim_kyc_flag	Curated	Customer Complia..	KYC completion a..	kyc_flag_code	kyc_flag_sk Onboa..
CA-FACT-001	Customer Activity ..	fact_customer_acti..	Curated	Customer x Day x ..	Daily activities acr..	activity_id	activity_sk OmniaE..
CA-FACT-002	Balance Snapshot ..	fact_account_bala..	Curated	Account x Day	EOD balances for ..	acct_id+date	snapshot_sk Core..
CA-FACT-003	Interaction Fact	fact_interaction	Curated	Interaction Event	Contact center and..	interaction_id	interaction_sk CX-..
CA-FACT-004	Marketing Respon..	fact_marketing_res..	Curated	Campaign x Custo..	Email/SMS/Push r..	campaign_id+cust	response_sk Mktg..
CA-BRD-001	Customer-Address..	bridge_customer_..	Curated	Customer x Address	Many-to-many ma..	customer_id+addr	cust_addr_sk CRM..
CA-BRD-002	Customer-Househ..	bridge_customer_..	Curated	Customer x House..	Membership mappi..	customer_id+hh	cust_hh_sk CRM-..
CA-BRD-003	Customer-Segmen..	bridge_customer_s..	Curated	Customer x Segm..	Many-to-many seg..	customer_id+seg	cust_seg_sk SegE..
CA-FACT-005	Lifecycle Event Fact	fact_customer_life..	Curated	Customer x Event	Milestone lifecycle ..	lifecycle_event_id	lifecycle_event_sk ..
CA-FACT-006	Support Case Fact	fact_customer_sup..	Curated	Case	Support case metri..	case_id	case_sk ServiceD..
CA-FACT-007	Digital Session Fact	fact_digital_session	Curated	Session	Web and app sessi..	session_id	session_sk Digital..
CA-DIM-011	Device Dimension	dim_device	Curated	Device	Device attributes f..	device_id	device_sk Digital3..
CA-DIM-012	Consent History	dim_consent_histo..	Curated	Consent (SCD2)	Historical consent ..	consent_id	consent_sk Conse..

Notes:

- CDC Type reflects the upstream capture marker used for change detection; no direct database logs are accessed from analytics layers.
- Late-Arriving Handling is restricted to controlled backfill windows and replay jobs scoped to business partitions to minimize operational impact.
- Stewardship teams confirm key changes via the monthly Schema Governance Forum; any entity with Review Status "Approved" has a valid control attestation within the current quarter.

Page 4 -- Field-Level Specifications: dim_customer and fact_customer_activity

This section details column-level specifications for two high-use entities. Definitions have been validated by Data Governance and DE-CA. PII categories reflect Meridian Capital's standard data classification: None, PII-Contact, and PII-Sensitive. Transformation references indicate logical mappings and standardization steps governed by Customer Analytics v5 (see mapping tickets under MC-ETLCA-10xxx). Defaults are applied within the curated layer only where specified for consistency and audit traceability.

Table: dim_customer (CA-DIM-001)

Table	Column	Data Type	Nullable	Default	Business Definiti..	Transformation L..	PII Category Qual..
dim_customer	customer_sk	BIGINT	N	Generated	Surrogate key uniq..	XFRM-CA-001	None DQ-CA-201 ..
dim_customer	customer_id	VARCHAR(64)	N	N/A	Natural key assign..	XFRM-CA-002	None DQ-CA-202 ..
dim_customer	customer_hash	CHAR(64)	N	N/A	Hash of selected id..	XFRM-CA-003	None DQ-CA-203 ..
dim_customer	household_sk	BIGINT	Y	NULL	Surrogate key of a..	XFRM-CA-004	None DQ-CA-204 ..
dim_customer	first_name	VARCHAR(128)	Y	NULL	Customer given na..	XFRM-CA-005	PII-Contact DQ-CA..
dim_customer	middle_name	VARCHAR(128)	Y	NULL	Middle name or init..	XFRM-CA-006	PII-Contact DQ-CA..
dim_customer	last_name	VARCHAR(128)	Y	NULL	Family name as m..	XFRM-CA-007	PII-Contact DQ-CA..
dim_customer	full_name	VARCHAR(384)	Y	NULL	Display name cons..	XFRM-CA-008	PII-Contact DQ-CA..
dim_customer	date_of_birth	DATE	Y	NULL	Date of birth for ag..	XFRM-CA-009	PII-Sensitive DQ-C..
dim_customer	age_band	VARCHAR(32)	Y	NULL	Derived age brack..	XFRM-CA-010	None DQ-CA-210 ..
dim_customer	gender	VARCHAR(32)	Y	NULL	Self-reported gend..	XFRM-CA-011	PII-Sensitive DQ-C..
dim_customer	email_address	VARCHAR(256)	Y	NULL	Primary email addr..	XFRM-CA-012	PII-Contact DQ-CA..
dim_customer	phone_mobile	VARCHAR(64)	Y	NULL	Mobile number in ..	XFRM-CA-013	PII-Contact DQ-CA..
dim_customer	phone_home	VARCHAR(64)	Y	NULL	Alternate landline ..	XFRM-CA-014	PII-Contact DQ-CA..
dim_customer	address_current_sk	BIGINT	Y	NULL	Surrogate key of c..	XFRM-CA-015	None DQ-CA-215 ..
dim_customer	country_code	CHAR(2)	Y	NULL	ISO 3166-1 alpha-..	XFRM-CA-016	None DQ-CA-216 ..
dim_customer	residency_status	VARCHAR(64)	Y	NULL	Residency categori..	XFRM-CA-017	None DQ-CA-217 ..
dim_customer	join_date	DATE	Y	NULL	First join date reco..	XFRM-CA-018	None DQ-CA-218 ..
dim_customer	tenure_months	INTEGER	Y	NULL	Months since join ..	XFRM-CA-019	None DQ-CA-219 ..
dim_customer	status_code	VARCHAR(32)	N	'ACTIVE'	Current lifecycle st..	XFRM-CA-020	None DQ-CA-220 ..
dim_customer	status_effective_fr..	TIMESTAMP	N	Current load time	Start timestamp for..	XFRM-CA-021	None DQ-CA-221 ..
dim_customer	status_effective_to	TIMESTAMP	Y	NULL	End timestamp for ..	XFRM-CA-022	None DQ-CA-222 ..
dim_customer	is_current	CHAR(1)	N	'Y'	Flag marking the a..	XFRM-CA-023	None DQ-CA-223 ..
dim_customer	kyc_completed_flag	CHAR(1)	Y	NULL	Indicates KYC com..	XFRM-CA-024	None DQ-CA-224 ..
dim_customer	marketing_opt_in_f..	CHAR(1)	Y	NULL	Indicates active pe..	XFRM-CA-025	None DQ-CA-225 ..
dim_customer	preferred_channel	VARCHAR(32)	Y	NULL	Preferred contact c..	XFRM-CA-026	None DQ-CA-226 ..
dim_customer	risk_rating_code	VARCHAR(16)	Y	NULL	Risk rating classific..	XFRM-CA-027	None DQ-CA-227 ..
dim_customer	vip_flag	CHAR(1)	Y	'N'	Indicates VIP hand..	XFRM-CA-028	None DQ-CA-228 ..
dim_customer	source_system_co..	VARCHAR(32)	N	'CRM'	Abbreviation of up..	XFRM-CA-029	None DQ-CA-229 ..
dim_customer	inserted_at_utc	TIMESTAMP	N	Current load time	Ingestion timestam..	XFRM-CA-030	None DQ-CA-230 ..
dim_customer	updated_at_utc	TIMESTAMP	N	Current load time	Last update timest..	XFRM-CA-031	None DQ-CA-231 ..
dim_customer	effective_from_utc	TIMESTAMP	N	Current load time	Start of SCD2 reco..	XFRM-CA-032	None DQ-CA-232 ..
dim_customer	effective_to_utc	TIMESTAMP	Y	NULL	End of SCD2 recor..	XFRM-CA-033	None DQ-CA-233 ..
dim_customer	record_version	INTEGER	N	1	Monotonic version ..	XFRM-CA-034	None DQ-CA-234 ..

Table: fact_customer_activity (CA-FACT-001)

Table	Column	Data Type	Nullable	Default	Business Definiti..	Transformation L..	PII Category Qual..
fact_customer_acti..	activity_sk	BIGINT	N	Generated	Surrogate key for t..	XFRM-CA-101	None DQ-CA-301 ..
fact_customer_acti..	activity_id	VARCHAR(64)	N	N/A	Upstream unique i..	XFRM-CA-102	None DQ-CA-302 ..
fact_customer_acti..	customer_sk	BIGINT	N	N/A	Foreign key to dim..	XFRM-CA-103	None DQ-CA-303 ..
fact_customer_acti..	activity_date	DATE	N	N/A	Calendar date of th..	XFRM-CA-104	None DQ-CA-304 ..
fact_customer_acti..	activity_ts_utc	TIMESTAMP	N	N/A	Exact UTC timesta..	XFRM-CA-105	None DQ-CA-305 ..

fact_customer_acti..	channel_sk	BIGINT	Y	NULL	Interaction channel..	XFRM-CA-106	None DQ-CA-306 ..
fact_customer_acti..	device_sk	BIGINT	Y	NULL	Device used for di..	XFRM-CA-107	None DQ-CA-307 ..
fact_customer_acti..	activity_type_code	VARCHAR(32)	N	N/A	Coded activity type..	XFRM-CA-108	None DQ-CA-308 ..
fact_customer_acti..	product_category_..	VARCHAR(32)	Y	NULL	Product category r..	XFRM-CA-109	None DQ-CA-309 ..
fact_customer_acti..	transaction_amount	DECIMAL(18,2)	Y	NULL	Monetary value as..	XFRM-CA-110	None DQ-CA-310 ..
fact_customer_acti..	currency_code	CHAR(3)	Y	NULL	ISO 4217 currency..	XFRM-CA-111	None DQ-CA-311 ..
fact_customer_acti..	fee_amount	DECIMAL(18,2)	Y	NULL	Fee charged at the..	XFRM-CA-112	None DQ-CA-312 ..
fact_customer_acti..	balance_after	DECIMAL(22,4)	Y	NULL	Account balance i..	XFRM-CA-113	None DQ-CA-313 ..
fact_customer_acti..	location_country_c..	CHAR(2)	Y	NULL	Country associate..	XFRM-CA-114	None DQ-CA-314 ..
fact_customer_acti..	session_id	VARCHAR(64)	Y	NULL	Digital session ide..	XFRM-CA-115	None DQ-CA-315 ..
fact_customer_acti..	marketing_campai..	VARCHAR(64)	Y	NULL	Campaign identifie..	XFRM-CA-116	None DQ-CA-316 ..
fact_customer_acti..	response_flag	CHAR(1)	Y	NULL	Indicates a measur..	XFRM-CA-117	None DQ-CA-317 ..
fact_customer_acti..	event_id	VARCHAR(64)	Y	NULL	External event refe..	XFRM-CA-118	None DQ-CA-318 ..
fact_customer_acti..	source_event_id	VARCHAR(64)	N	N/A	Upstream messag..	XFRM-CA-119	None DQ-CA-319 ..
fact_customer_acti..	source_system_co..	VARCHAR(32)	N	'EVT'	Abbreviation of ev..	XFRM-CA-120	None DQ-CA-320 ..
fact_customer_acti..	is_reversed_flag	CHAR(1)	Y	'N'	Indicates whether t..	XFRM-CA-121	None DQ-CA-321 ..
fact_customer_acti..	reversal_event_id	VARCHAR(64)	Y	NULL	Identifier of the rev..	XFRM-CA-122	None DQ-CA-322 ..
fact_customer_acti..	load_batch_id	VARCHAR(64)	N	N/A	Batch identifier for ..	XFRM-CA-123	None DQ-CA-323 ..
fact_customer_acti..	inserted_at_utc	TIMESTAMP	N	Current load time	Ingestion timestam..	XFRM-CA-124	None DQ-CA-324 ..
fact_customer_acti..	updated_at_utc	TIMESTAMP	N	Current load time	Last update timest..	XFRM-CA-125	None DQ-CA-325 ..
fact_customer_acti..	dwh_partition_date	DATE	N	activity_date	Partitioning key for..	XFRM-CA-126	None DQ-CA-326 ..

Additional Notes:

- All VARCHAR fields are UTF-8 encoded; trimming rules remove trailing/leading whitespace while preserving meaningful internal spaces.
- DQ-CA-2xx/3xx rulesets cover mandatory presence, domain conformance, and referential integrity checks. Failures route to the Data Quality Office queue with ticket prefix DQO-CA.
- Ownership reflects accountable team for business meaning; DE-CA retains technical stewardship for data model integrity and refresh cadence.

ETL Schema Specification -- Customer Analytics v5

Meridian Capital

Pages 5-6

Page 5 -- Dimension: Customer (extended attributes), Reference Data, and Lineage Summaries

This page enumerates the extended attribute set for dim_customer as of v5.2, describing data types, SCD handling, and concise business definitions suitable for Customer Analytics workloads. It also captures the current reference code set for customer segmentation and a point-in-time lineage snapshot for priority attributes. Reviewed by Customer Analytics Engineering and Data Platform Operations on 2026-07-15T09:30:00Z per ticket DATA-7421.

dim_customer column specification (continued, v5.2, schema_id: CA.DIM.CUSTOMER.V5)

Column Name	Data Type	Nullable	Source System	Transform Rule Id	Default	SCD Type	Business Definiti..
loyalty_tier_code	STRING(16)	N	CRM_CORE	TR-DC-0527	'BASIC'	2	Current loyalty tier ..
marketing_opt_in_f..	BOOLEAN	N	MKT_HUB	TR-DC-0531	FALSE	2	Indicates customer..
email_hash_sha256	STRING(64)	Y	MKT_HUB	TR-DC-0543	NULL	1	Irreversible hash of..
phone_e164	STRING(20)	Y	CRM_CORE	TR-DC-0549	NULL	2	Normalized phone ..
birth_year	INT32	Y	CRM_CORE	TR-DC-0554	NULL	1	Four-digit year of b..
household_id	STRING(40)	Y	CRM_CORE	TR-DC-0557	NULL	2	Derived household..
preferred_languag..	STRING(8)	Y	CRM_CORE	TR-DC-0560	'en'	2	Two-letter languag..
country_iso2	STRING(2)	N	CRM_CORE	TR-DC-0564	'US'	2	Customer country i..
region_code	STRING(8)	Y	CRM_CORE	TR-DC-0567	NULL	2	State/province/regi..
city	STRING(120)	Y	CRM_CORE	TR-DC-0569	NULL	2	City name from ma..
postal_code	STRING(16)	Y	CRM_CORE	TR-DC-0573	NULL	2	Postal or ZIP code ..
address_quality_s..	DECIMAL(5,2)	Y	CRM_CORE	TR-DC-0578	0.00	1	Address validation ..
first_purchase_date	DATE	Y	PAYMENTS_LED..	TR-DC-0581	NULL	1	Date of first record..
last_purchase_date	DATE	Y	PAYMENTS_LED..	TR-DC-0585	NULL	1	Most recent purch..
total_lifetime_value	DECIMAL(18,2)	Y	PAYMENTS_LED..	TR-DC-0589	0.00	1	Sum of net revenu..
avg_order_value	DECIMAL(18,2)	Y	PAYMENTS_LED..	TR-DC-0593	0.00	1	Average order valu..
churn_risk_score	DECIMAL(5,4)	Y	MODEL_SERVING	TR-DC-0597	0.0000	1	Probability score fr..
engagement_scor..	INT32	Y	EVENT_STREAM	TR-DC-0601	0	1	Interaction score o..
kyc_status_code	STRING(16)	Y	RISK_KYC	TR-DC-0604	'UNKNOWN'	2	Overall KYC status..
aml_review_flag	BOOLEAN	Y	RISK_KYC	TR-DC-0607	FALSE	2	Flag indicating AM..
consent_gdpr_flag	BOOLEAN	Y	MKT_HUB	TR-DC-0610	FALSE	2	EU data subject co..
consent_ccpa_flag	BOOLEAN	Y	MKT_HUB	TR-DC-0613	FALSE	2	California consum..
data_retention_cla..	STRING(16)	Y	DATA_GOV	TR-DC-0616	'STD'	1	Retention classific..
do_not_contact_flag	BOOLEAN	Y	MKT_HUB	TR-DC-0619	FALSE	2	Suppression indica..
vip_flag	BOOLEAN	Y	CRM_CORE	TR-DC-0622	FALSE	2	VIP designation for..
account_manager..	STRING(32)	Y	CRM_CORE	TR-DC-0626	NULL	2	Assigned account ..
referral_source_co..	STRING(16)	Y	CRM_CORE	TR-DC-0629	'UNKNOWN'	2	How the customer ..
acquisition_chann..	STRING(16)	Y	MKT_HUB	TR-DC-0632	'UNKNOWN'	2	Primary marketing ..
device_preference	STRING(16)	Y	EVENT_STREAM	TR-DC-0635	'UNKNOWN'	1	Most-used device ..
time_zone	STRING(64)	Y	CRM_CORE	TR-DC-0638	'UTC'	2	Preferred time zon..
create_ts_utc	TIMESTAMP	N	ETL	TR-DC-0641	CURRENT_TIMES..	1	Record creation ti..
update_ts_utc	TIMESTAMP	N	ETL	TR-DC-0644	CURRENT_TIMES..	1	Last update timest..
record_status_code	STRING(8)	N	ETL	TR-DC-0647	'ACTIVE'	1	Lifecycle status of t..
dedupe_group_id	STRING(40)	Y	ETL	TR-DC-0650	NULL	1	Identifier grouping ..
merge_master_cu..	STRING(40)	Y	ETL	TR-DC-0653	NULL	1	Canonical custome..

Reference data -- customer_segment_code (active code set, ref_id: CA.REF.SEGMENT.V3, validated 2026-07-12 per REF-3189)

Code	Description	Effective From (U..	Effective To (UTC)	Source	Owner	Ticket Ref

RETAIN	Retention priority c..	2025-11-01T00:00..	9999-12-31T23:59..	MKT_HUB	Growth Analytics	REF-3189
ACQ_NEW	New acquisition (<..	2025-11-01T00:00..	9999-12-31T23:59..	MKT_HUB	Growth Analytics	REF-3189
VIP	High-value VIP co..	2025-11-01T00:00..	9999-12-31T23:59..	CRM_CORE	Customer Success..	REF-3192
DORMANT	No activity >=180 ..	2025-11-01T00:00..	9999-12-31T23:59..	EVENT_STREAM	Cust Analytics Ops	REF-3201
RISK_REVIEW	Under risk review	2025-11-01T00:00..	9999-12-31T23:59..	RISK_KYC	Risk & Compliance..	REF-3220
LOYAL_GOLD	Loyalty tier Gold	2025-11-01T00:00..	9999-12-31T23:59..	CRM_CORE	Cust Analytics Ops	REF-3189
LOYAL_PLAT	Loyalty tier Platinum	2025-11-01T00:00..	9999-12-31T23:59..	CRM_CORE	Cust Analytics Ops	REF-3189
TEST_EXCL	Test/control exclus..	2025-11-01T00:00..	9999-12-31T23:59..	MKT_HUB	Experimentation W..	REF-3215

Lineage snapshot -- selected attributes (captured 2026-07-10T22:00:00Z, audit batch: LG-CA-2207, QA ticket DQ-5610)

Target Field	Upstream Source..	Extraction Job Id	Transform Step	Load Job Id	Last Validated (U..	QA Status	Evidence Doc Id
loyalty_tier_code	crm.customer.loyal..	CRM_CORE_FUL..	TR-DC-0527	CA_DIMCUST_SC..	2026-07-10T22:15:..	PASS	EVD-CA-11492
marketing_opt_in_f..	mkt.customer.opt_in	MKT_HUB_DELT..	TR-DC-0531	CA_DIMCUST_SC..	2026-07-10T22:15:..	PASS	EVD-CA-11493
email_hash_sha256	mkt.customer.email	MKT_HUB_DELT..	TR-DC-0543	CA_DIMCUST_TY..	2026-07-10T22:16:..	PASS	EVD-CA-11494
phone_e164	crm.contact.phone	CRM_CORE_FUL..	TR-DC-0549	CA_DIMCUST_SC..	2026-07-10T22:18:..	PASS	EVD-CA-11495
total_lifetime_value	pay.order.net_amo..	PAY_LEDGER_D..	TR-DC-0589	CA_DIMCUST_TY..	2026-07-10T22:22:..	PASS	EVD-CA-11496
churn_risk_score	model.churn.score	MODEL_SCORES..	TR-DC-0597	CA_DIMCUST_TY..	2026-07-10T22:23:..	PASS	EVD-CA-11497
kyc_status_code	risk.kyc.status	RISK_KYC_SYNC..	TR-DC-0604	CA_DIMCUST_SC..	2026-07-10T22:25:..	PASS	EVD-CA-11498
consent_gdpr_flag	mkt.consent.gdpr	MKT_HUB_DELT..	TR-DC-0610	CA_DIMCUST_SC..	2026-07-10T22:27:..	PASS	EVD-CA-11499

Page 6 -- Batch Dependencies, SLAs, and Data Quality Controls

This page summarizes operational scheduling, upstream dependencies, and service levels for the dim_customer pipeline, alongside mandatory data quality controls, rolling metrics, and v5.x change history. The schedule reflects the 2026-07 release calendar approved by Data Platform Operations per CAB-2026-07-09-CA.

ETL job schedule and SLA -- Customer Analytics v5 (calendar effective 2026-07-11)

Job Id	Job Name	Frequency	Window (UTC)	SLA (UTC)	Upstream Depen..	Retry Policy Id	Owning Team On..
CA_DIMCUST_EX..	Extract CRM core ..	Daily	21:00-21:25	22:00	CRM_CORE_FUL..	RP-STD-3x	Data Platform Ops ..
CA_DIMCUST_EX..	Extract marketing ..	Daily	21:05-21:20	22:00	MKT_HUB_DELT..	RP-STD-3x	Customer Analytic..
CA_DIMCUST_EX..	Extract payments l..	Daily	21:10-21:30	22:05	PAY_LEDGER_D..	RP-STD-3x	Revenue Data Eng..
CA_DIMCUST_EX..	Sync risk/KYC attri..	Daily	21:00-21:15	22:00	RISK_KYC_SYNC..	RP-STD-3x	Risk Data Services..
CA_DIMCUST_TY..	Apply Type 1 attrib..	Daily	21:30-21:55	22:10	CA_DIMCUST_EX..	RP-STD-2x	Customer Analytic..
CA_DIMCUST_SC..	Apply Type 2 SCD ..	Daily	21:55-22:20	22:30	CA_DIMCUST_TY..	RP-STD-2x	Customer Analytic..
CA_DIMCUST_D..	Run DQ controls	Daily	22:20-22:35	22:40	CA_DIMCUST_SC..	RP-STD-1x	Data Quality Office..
CA_DIMCUST_LI..	Capture lineage sn..	Daily	22:35-22:40	22:45	CA_DIMCUST_SC..	RP-STD-1x	Data Platform Ops ..
CA_REF_SEGME..	Refresh segment c..	Weekly Sun	20:40-20:50	21:10	MKT_HUB_DELT..	RP-STD-2x	Growth Analytics g..
CA_EXPORT_AN..	Publish curated ou..	Daily	22:40-22:55	23:10	CA_DIMCUST_D..	RP-STD-2x	Analytics Product ..
CA_BACKFILL_DI..	Historical backfill r..	Ad hoc	As scheduled	As defined	As defined	RP-APP-approval	Data Platform Ops ..
CA_AUDIT_SNAP..	Compliance audit c..	Monthly 1st	01:00-01:20	01:40	CA_EXPORT_AN..	RP-STD-1x	Data Governance ..

Data quality controls -- dim_customer (control pack: DQ-CA-DIMCUST-V5.2, last full pass 2026-07-14T22:33:00Z per DQ-5610)

Control Id	Control Type	Target Field(s)	Threshold	Sample Size	Status	Last Run (UTC)	Next Run (UTC) T..
DQ-CUST-001	Uniqueness	customer_key	0 duplicates	100%	PASS	2026-07-14T22:25:..	2026-07-15T22:25:..
DQ-CUST-002	Not-null	country_iso2	<=0.10% null	100%	PASS	2026-07-14T22:25:..	2026-07-15T22:25:..
DQ-CUST-003	Domain check	loyalty_tier_code	<=0.05% out-of-set	100%	PASS	2026-07-14T22:26:..	2026-07-15T22:26:..
DQ-CUST-004	Range check	churn_risk_score	<=0.01% outside [..	100%	PASS	2026-07-14T22:26:..	2026-07-15T22:26:..
DQ-CUST-005	Freshness	update_ts_utc	<=60 min lag p95	100%	PASS	2026-07-14T22:27:..	2026-07-15T22:27:..
DQ-CUST-006	Referential integrity	country_iso2 -> di..	<=0.10% orphan	100%	PASS	2026-07-14T22:27:..	2026-07-15T22:27:..
DQ-CUST-007	Pattern	phone_e164	<=0.50% non-E164	100%	PASS	2026-07-14T22:27:..	2026-07-15T22:27:..
DQ-CUST-008	Consistency	total_lifetime_valu..	<=0.01% violations	100%	PASS	2026-07-14T22:28:..	2026-07-15T22:28:..
DQ-CUST-009	Temporal	first_purchase_dat..	<=0.05% violations	100%	PASS	2026-07-14T22:28:..	2026-07-15T22:28:..
DQ-CUST-010	Conformance	kyc_status_code in..	<=0.10% out-of-set	100%	PASS	2026-07-14T22:29:..	2026-07-15T22:29:..
DQ-CUST-011	Completeness	marketing_opt_in_f..	<=0.50% gaps	100%	PASS	2026-07-14T22:29:..	2026-07-15T22:29:..
DQ-CUST-012	SCD validity	No overlapping SC..	0 overlaps	100%	PASS	2026-07-14T22:30:..	2026-07-15T22:30:..

Operational metrics (rolling 30 days ending 2026-07-14; owner: Data Platform Ops; report id: OPS-CA-DIMCUST-30D)

Metric	Target	P50	P95	Min	Max	Last Breach Date ..	Notes
Daily rows loaded	>= 18.0M	18.6M	19.1M	17.8M	19.4M	2026-07-03T22:45:..	Minor dip due to C..
Load duration (Typ..	<= 25 min	18m 42s	23m 55s	17m 58s	24m 31s	N/A	Within SLA.
Load duration (SC..	<= 25 min	19m 13s	24m 47s	18m 40s	25m 02s	2026-07-02T22:33:..	One run slightly ex..
DQ exception rate	<= 0.20%	0.06%	0.14%	0.03%	0.17%	N/A	Stable.
Late arrivals (recor..	<= 0.50%	0.22%	0.41%	0.16%	0.48%	N/A	Within target.
SCD2 changes/day	Informational	1.2M	1.6M	0.9M	1.8M	N/A	Driven by marketin..
Retry count per job	<= 2	0	1	0	1	N/A	No multi-retry even..
Export publication t..	<= 23:10 UTC	22:54	23:06	22:49	23:08	N/A	Meets publication ..
Freshness lag p95	<= 60 min	38m	52m	31m	59m	N/A	Meets freshness ta..
On-call pages	0-2 / month	0	1	0	1	2026-07-02T22:34:..	Single transient ale..

Change log -- Customer Analytics v5.x (CAB-approved items only)

Date (UTC)	Version	Change Ticket	Summary	Backward Compa..	Rollback Plan Id	Owner
2026-07-12T10:15..	v5.2	DATA-7421	Added consent_cc..	Yes	RB-CA-2026-07-12	Customer Analytic..
2026-06-09T11:00..	v5.1	DATA-7194	Introduced churn_r..	Yes	RB-CA-2026-06-09	Data Science Ops
2026-05-03T09:20..	v5.0.2	DATA-7038	Adjusted total_lifeti..	Yes	RB-CA-2026-05-03	Data Platform Ops
2026-04-17T13:40..	v5.0.1	DATA-6975	Introduced addres..	Yes	RB-CA-2026-04-17	Geo Data Steward
2026-03-28T08:55..	v5.0	DATA-6890	GA release of dim..	Yes	RB-CA-2026-03-28	Customer Analytic..
2026-02-14T15:05..	v4.9.9	DATA-6710	Pre-v5 staging: ad..	Yes	RB-CA-2026-02-14	Privacy Office Data
2026-01-09T12:30..	v4.9.8	DATA-6601	Introduced device_..	Yes	RB-CA-2026-01-09	Customer Engage..
2025-12-05T07:50..	v4.9.7	DATA-6488	Initial integration of..	Yes	RB-CA-2025-12-05	Risk & Compliance..

End of Pages 5-6.

Page 7 -- DimCustomer Field Catalog and Transform Mapping (Customer Analytics v5)

The following catalog enumerates the DimCustomer schema as implemented in Customer Analytics v5 for Meridian Capital. Field definitions reflect RFC CA-318 and Change Request CR-2026-Q2-09 approved by the Customer Analytics Governance Council on 2026-06-21. Transform Rule IDs (TR-CA-xxx) reference the canonical mapping library maintained by the Customer Analytics ETL Team; Data Quality Rule IDs (DQ-CA-xxx) are administered by the Data Quality Office. All timestamps are UTC. Review tickets: MC-ETL-4821, MC-DQ-2145, MC-MDM-1079.

Field Identifier	Data Type (DW)	Nullable	Source System	Source Object.Fie..	Transform Rule ID	Derivation / Defau..	Business Semant..
customer_key	BIGINT	No	Generated	N/A	TR-CA-100	DW sequence	Surrogate primary ..
customer_id	VARCHAR(64)	No	CRM	crm.customer.id	TR-CA-121	Trim, upper, dedupe	External natural ke..
customer_guid	CHAR(36)	Yes	MDM	mdm.party.guid	TR-CA-204	Validate UUID	Cross-system iden..
customer_status	VARCHAR(16)	No	CRM	crm.customer.statu..	TR-CA-203	Map code to canon..	Lifecycle status AC..
first_name	VARCHAR(80)	Yes	CRM	crm.customer.first_..	TR-CA-130	Title case	Given name Alicia ..
last_name	VARCHAR(120)	Yes	CRM	crm.customer.last_..	TR-CA-131	Title case	Family name Moral..
full_name	VARCHAR(160)	Yes	Derived	first_name + last_n..	TR-CA-135	Concatenate, trim	Display name Alici..
email_address	VARCHAR(254)	Yes	CRM	crm.contact.email	TR-CA-140	Lower, format check	Primary email a.m..
phone_e164	VARCHAR(20)	Yes	CallCenter	cc.customer.phone	TR-CA-146	Normalize E.164	Primary phone +13..
birth_date	DATE	Yes	CoreBanking	core.party.birth_dt	TR-CA-150	Validate range	Date of birth 1987-..
country_iso2	CHAR(2)	Yes	MDM	mdm.address.coun..	TR-CA-210	ISO 3166-1 alpha-2	Country code US ..
state_province	VARCHAR(64)	Yes	MDM	mdm.address.regi..	TR-CA-211	Standardize region	State or province ..
city	VARCHAR(120)	Yes	MDM	mdm.address.city	TR-CA-212	Proper case	City San Francisco..
postal_code	VARCHAR(20)	Yes	MDM	mdm.address.post..	TR-CA-213	Normalize	Postal or ZIP 9410..
street_line1	VARCHAR(160)	Yes	MDM	mdm.address.line1	TR-CA-214	Trim	Address line 1 425..
street_line2	VARCHAR(160)	Yes	MDM	mdm.address.line2	TR-CA-214	Trim	Address line 2 Suit..
segment_code	VARCHAR(16)	Yes	MDM	mdm.segment.code	TR-CA-230	Map to canonical	Customer segment..
preferred_language	CHAR(2)	Yes	CRM	crm.preferences.la..	TR-CA-175	ISO 639-1	Locale preference ..
signup_channel	VARCHAR(32)	Yes	Digital	web.signup.channel	TR-CA-260	Canonical map	Acquisition channe..
signup_timestamp..	TIMESTAMP	Yes	Digital	web.signup.ts	TR-CA-261	Parse, to UTC	First signup time 2..
last_activity_timest..	TIMESTAMP	Yes	Multi	usage.last_seen_ts	TR-CA-262	Max over sources	Recent activity 202..
loyalty_tier	VARCHAR(12)	Yes	Loyalty	loy.member.tier	TR-CA-305	Canonical map	Rewards tier GOL..
opt_in_marketing	BOOLEAN	Yes	Marketing	mkt.preference.opt..	TR-CA-176	Coalesce false	Marketing consent ..
gdpr_consent_time..	TIMESTAMP	Yes	Marketing	mkt.preference.gd..	TR-CA-177	Nullable non-EU	Consent capture ti..
risk_rating	DECIMAL(5,2)	Yes	Risk	risk.profile.rating	TR-CA-410	Bound 0-10	Internal risk score ..
kyc_status	VARCHAR(16)	Yes	Compliance	comp.kyc.status	TR-CA-411	Canonical map	KYC completion fla..
account_open_date	DATE	Yes	CoreBanking	core.account.open..	TR-CA-151	Validate not future	First account open ..
account_close_date	DATE	Yes	CoreBanking	core.account.close..	TR-CA-152	Null if active	Final account clos..
is_active	BOOLEAN	No	Derived	status, close_dt	TR-CA-155	status <> CLOSED	Active indicator tru..
record_effective_ts	TIMESTAMP	No	ETL	load.batch_ts	TR-CA-001	Load timestamp	SCD2 effective sta..
record_expiry_ts	TIMESTAMP	Yes	ETL	N/A	TR-CA-002	Null for current	SCD2 effective en..
current_record_flag	BOOLEAN	No	ETL	N/A	TR-CA-003	True if expiry null	SCD2 current flag t..
load_batch_id	BIGINT	No	ETL	etl.control.batch_id	TR-CA-010	From control plane	Ingestion batch ide..
source_system_pri..	SMALLINT	Yes	ETL	merge.conflict.prior..	TR-CA-270	1=MDM,2=CRM,3..	Conflict resolution t..

Notes:

- Data custody: Primary stewardship resides with the Customer Analytics ETL Team; domain-specific attributes remain under respective source stewards as listed above.
- Approved lineage review completed 2026-06-29 under MC-LIN-773; no variances from TR-CA specifications were found. Next review window: 2026-Q3 cycle.

Page 8 -- FactCustomerInteraction Field Catalog and Monitoring KPIs (Customer Analytics v5)

This page details the FactCustomerInteraction event grain schema and the operational monitoring KPIs associated with Customer Analytics v5. The event table is at event-level grain (one row per interaction per customer), validated in MC-ETL-4930 and MC-DQ-2172. Ownership is shared between the Customer Analytics ETL Team and Digital Products. KPI thresholds and targets were ratified by the Data Quality Office on 2026-06-20.

Field Identifier	Data Type (DW)	Nullable	Grain/Key Role	Source System	Source Object.Fie..	Transform Rule ID	Business Semant..
event_key	BIGINT	No	Surrogate PK	Generated	N/A	TR-CA-500	Unique row identifi..
event_guid	CHAR(36)	Yes	Cross-system ID	Digital	web.event.guid	TR-CA-512	Source event UUI..
customer_key	BIGINT	No	FK to DimCustomer	ETL	dim_customer.cust..	TR-CA-501	Dimension foreign ..
event_type_code	VARCHAR(24)	No	Fact attribute	Multi	src.event.type_code	TR-CA-520	Canonical event ty..
channel_code	VARCHAR(16)	Yes	Fact attribute	Multi	src.event.channel	TR-CA-521	Interaction channel..
event_timestamp_..	TIMESTAMP	No	Event time	Digital	src.event.ts	TR-CA-510	Parsed, UTC-norm..
session_id	VARCHAR(64)	Yes	Session grouping	Digital	web.session.id	TR-CA-523	Browser/app sessi..
campaign_id	VARCHAR(64)	Yes	MKT attribution	Marketing	mkt.campaign.id	TR-CA-540	Campaign referenc..
product_id	VARCHAR(64)	Yes	Product ref	Catalog	cat.product.id	TR-CA-545	Referenced produc..
geo_country_iso2	CHAR(2)	Yes	Geo attribute	Digital	web.event.country	TR-CA-532	Derived from IP/ge..
device_type	VARCHAR(16)	Yes	Device attribute	Digital	web.event.device	TR-CA-533	Canonical device c..
referrer	VARCHAR(255)	Yes	Context	Digital	web.event.referrer	TR-CA-534	Standardized URL ..
source_system	VARCHAR(24)	No	Lineage	ETL	ingest.source_name	TR-CA-505	Originating pipelin..
load_batch_id	BIGINT	No	Operational	ETL	control.batch_id	TR-CA-506	Ingestion batch ide..
ingestion_timesta..	TIMESTAMP	No	Operational	ETL	control.ingest_ts	TR-CA-507	Raw ingest time 20..
processing_timest..	TIMESTAMP	No	Operational	ETL	control.proc_ts	TR-CA-508	Warehouse load ti..
attribution_model_..	VARCHAR(16)	Yes	Attribution	Analytics	an.lt.version	TR-CA-560	Model version tag ..
revenue_amount	DECIMAL(18,2)	Yes	Measure	Billing	bill.event.amount	TR-CA-570	Currency-aligned v..
currency_iso3	CHAR(3)	Yes	Measure context	Billing	bill.event.currency	TR-CA-571	ISO 4217 USD Fin..
is_test_event	BOOLEAN	Yes	QA filtering	Digital	web.event.test_flag	TR-CA-590	Exclude from KPIs ..
delete_flag	BOOLEAN	No	Soft delete	ETL	control.delete_flag	TR-CA-599	True if retracted fal..

Operational Data Quality KPIs -- Customer Analytics v5 (MC-DQ-2172)

KPI ID	Metric Name	Definition	Target	Threshold	Window	Source Table	Owner Team JIR..
KPI-CA-01	Event Timeliness	Ingest-to-warehou..	<= 15 minutes	20 minutes	Rolling 24h	fact_customer_inte..	Customer Analytic..
KPI-CA-02	Join Conformity	Orphaned events ..	<= 0.10%	0.25%	Daily	fact vs. dim join	Data Quality Office..
KPI-CA-03	Duplicate Event R..	Duplicate event_g..	<= 0.05%	0.10%	Daily	fact_customer_inte..	Digital Products M..
KPI-CA-04	Currency Presence	Percent of revenue..	>= 99.80%	99.50%	Weekly	fact_customer_inte..	Finance Data MC-..
KPI-CA-05	Email Format Valid..	Valid email_addres..	>= 98.50%	98.00%	Weekly	dim_customer	Marketing Tech Op..
KPI-CA-06	Geolocation Compl..	Non-null geo_coun..	>= 97.00%	96.50%	Rolling 7d	fact_customer_inte..	Digital Products M..
KPI-CA-07	Soft Delete Adhere..	Rows flagged delet..	100%	99.50%	Rolling 7d	fact_customer_inte..	Customer Analytic..
KPI-CA-08	Risk Rating Range	risk_rating within 0..	100%	99.95%	Monthly	dim_customer	Risk Analytics MC-..

Notes and Findings:

- June 2026 monitoring window closed without SLA breaches; all KPIs remained Green. Executive summary filed under MC-REP-3307 on 2026-06-30.
- Minor data drift observed in channel_code distribution (WEB vs. APP) during 2026-06-22-2026-06-24; root cause tracked to A/B traffic split change (MC-DIG-1451). No schema or transform updates required.
- Next KPI calibration review scheduled for 2026-07-22; ownership: Data Quality Office with participation from Customer Analytics ETL and Digital Products.

ETL Schema Specification -- Customer Analytics v5 -- Meridian Capital

The following section documents field-level validation coverage and observed data quality outcomes for Customer Analytics v5. Findings reflect the most recent full-cycle execution completed by Customer Analytics Engineering (CAE) on 2026-07-18T03:14:27Z with oversight from Enterprise Data Quality (EDQ). Thresholds align to the Customer Data Governance Council-approved tolerances under DQ Standard MRC-DQ-CA-005, and remediation items are tracked in Jira and ServiceNow as referenced.

Field Name	Table	Data Type	Nullable	Business Rule Su..	Check Type	Threshold (Max)	Last Check (UTC)..
customer_key	customer_dim_v5	BIGINT	No	Unique surrogate k..	Uniqueness/Null	0 duplicates/0 nulls	2026-07-18T03:12:..
source_system	customer_dim_v5	VARCHAR(32)	No	Value in approved ..	Domain	0.10% violations	2026-07-18T03:12:..
crm_customer_id	customer_dim_v5	VARCHAR(64)	Yes	If source_system=..	Conditional Null	0.50% conditional ..	2026-07-18T03:12:..
household_id	customer_dim_v5	VARCHAR(64)	Yes	If present, must joi..	Referential	0.25% unresolvable	2026-07-18T03:12:..
segmentation_code	customer_dim_v5	VARCHAR(16)	Yes	If present, value in ..	Domain	0.25% violations	2026-07-18T03:12:..
email_hash	customer_dim_v5	CHAR(64)	Yes	If email_normalize..	Format/Length	0.10% invalid	2026-07-18T03:12:..
email_normalized	customer_dim_v5	VARCHAR(254)	Yes	If present, must co..	Format	0.50% invalid	2026-07-18T03:12:..
phone_e164	customer_dim_v5	VARCHAR(20)	Yes	If present, must be..	Format	0.50% invalid	2026-07-18T03:12:..
birth_date	customer_dim_v5	DATE	Yes	Must not be in futu..	Range/Temporal	0.10% out-of-range	2026-07-18T03:12:..
kyc_status	customer_dim_v5	VARCHAR(16)	No	Value in {PENDIN..	Domain	0.10% violations	2026-07-18T03:12:..
marketing_opt_in	customer_dim_v5	BOOLEAN	Yes	If null, default inter..	Null/Default	2.00% null rate	2026-07-18T03:12:..
country_code	customer_dim_v5	CHAR(2)	Yes	If present, must be..	Domain	0.25% invalid	2026-07-18T03:12:..
region_code	customer_dim_v5	VARCHAR(8)	Yes	If present, must joi..	Referential	0.50% unresolvable	2026-07-18T03:12:..
address_line1	customer_dim_v5	VARCHAR(128)	Yes	Free text; if addres..	Dependency	0.25% violations	2026-07-18T03:12:..
postal_code	customer_dim_v5	VARCHAR(16)	Yes	If present, must co..	Format	0.50% invalid	2026-07-18T03:12:..
created_at	customer_dim_v5	TIMESTAMP	No	Must be <= current..	Temporal	0.05% future	2026-07-18T03:12:..
updated_at	customer_dim_v5	TIMESTAMP	Yes	Must be >= create..	Temporal	0.25% inversion	2026-07-18T03:12:..
loyalty_tier	customer_dim_v5	VARCHAR(16)	Yes	If present, value in ..	Domain	0.25% violations	2026-07-18T03:12:..
preferred_language	customer_dim_v5	VARCHAR(8)	Yes	If present, must be..	Domain	0.25% invalid	2026-07-18T03:12:..
consent_updated_at	customer_dim_v5	TIMESTAMP	Yes	If marketing_opt_in..	Conditional Null	0.50% missing	2026-07-18T03:12:..
risk_score	customer_dim_v5	SMALLINT	Yes	If present, range 0-..	Range	0.25% out-of-range	2026-07-18T03:12:..
lifetime_value	customer_fact_v5	DECIMAL(18,2)	Yes	Non-negative; if ne..	Range/Flag	0.10% negative	2026-07-18T03:13:..
monthly_spend	customer_fact_v5	DECIMAL(18,2)	Yes	Non-negative; sum..	Range/Reconcile	0.50% variance	2026-07-18T03:13:..
txn_count_12mo	customer_fact_v5	INTEGER	Yes	Must be integer >=..	Range/Dependency	0.25% invalid	2026-07-18T03:13:..
first_txn_date	customer_fact_v5	DATE	Yes	If present, must be..	Temporal	0.25% inversion	2026-07-18T03:13:..
last_txn_date	customer_fact_v5	DATE	Yes	Must be <= as-of-d..	Temporal	0.05% future	2026-07-18T03:13:..
currency_code	customer_fact_v5	CHAR(3)	No	Value in ISO 4217 ..	Domain/Default	0.10% invalid	2026-07-18T03:13:..
asof_date	customer_fact_v5	DATE	No	Partition key; must ..	Partition	0.00% mismatch	2026-07-18T03:13:..
customer_key	customer_fact_v5	BIGINT	No	Must join to custo..	Referential	0.10% unresolvable	2026-07-18T03:13:..
channel_code	customer_fact_v5	VARCHAR(16)	Yes	If present, value in ..	Domain	0.50% violations	2026-07-18T03:13:..
acquisition_source	customer_dim_v5	VARCHAR(32)	Yes	If present, value in ..	Domain	0.50% violations	2026-07-18T03:12:..
geo_latitude	customer_dim_v5	DECIMAL(9,6)	Yes	If present, range -9..	Range	0.10% out-of-range	2026-07-18T03:12:..
geo_longitude	customer_dim_v5	DECIMAL(9,6)	Yes	If present, range -1..	Range	0.10% out-of-range	2026-07-18T03:12:..
churn_risk_band	customer_fact_v5	VARCHAR(8)	Yes	If present, value in ..	Domain	0.25% violations	2026-07-18T03:13:..
promo_eligibility	customer_dim_v5	BOOLEAN	Yes	If true, must have c..	Temporal/Conditio..	0.50% stale	2026-07-18T03:12:..
do_not_contact_flag	customer_dim_v5	BOOLEAN	No	If true, marketing_..	Cross-Field	0.10% conflict	2026-07-18T03:12:..
tenure_days	customer_fact_v5	INTEGER	Yes	If present, comput..	Derivation/Range	0.10% negative	2026-07-18T03:13:..
portfolio_segment	customer_fact_v5	VARCHAR(16)	Yes	If present, value in ..	Domain	0.25% violations	2026-07-18T03:13:..
credit_limit_bucket	customer_fact_v5	VARCHAR(16)	Yes	If present, must joi..	Referential	0.25% unresolvable	2026-07-18T03:13:..
csat_score	customer_fact_v5	DECIMAL(3,2)	Yes	If present, range 0...	Range	0.25% out-of-range	2026-07-18T03:13:..
referral_code	customer_dim_v5	VARCHAR(32)	Yes	If present, only upp..	Format/Length	0.50% invalid	2026-07-18T03:12:..
gdpr_region_flag	customer_dim_v5	BOOLEAN	Yes	If true, country_co..	Cross-Field	0.25% mismatch	2026-07-18T03:12:..
preferred_channel	customer_dim_v5	VARCHAR(16)	Yes	If present, must be..	Domain	0.50% violations	2026-07-18T03:12:..

Notes:

- Where warnings are indicated, tolerances remain within governance-approved limits; ongoing monitoring is active. Remediation items MRC-ETL-4217 and MRC-ETL-4329 are scheduled for completion within Sprint

CAE-2026.30 (target 2026-07-31) under release train CA-RT-5.12.

- All checks executed via EDQ framework batch 2026-07-18_B3; artifacts stored in mc_dataops.edq_audit_v2 with run_id=EDQ-CA-20260718-03.

ETL Schema Specification -- Customer Analytics v5 -- Meridian Capital

This section summarizes the operational load schedule and recent performance characteristics for Customer Analytics v5 pipelines. The scheduling matrix is maintained by SRE-Data with inputs from CAE, and the performance metrics reflect the trailing 14-day window concluding on 2026-07-18. SLAs and windows are aligned to business reporting cutoffs and upstream source availability. References to tickets denote tracked adjustments, routine maintenance, or post-incident reviews.

ETL Job Scheduling Matrix (Active as of 2026-07-18)

Job ID	Job Name	Source System	Target Dataset	Frequency	Window (UTC)	SLA (UTC)	Last 7-Day Succe..
ETL.CUST.DIM.L..	Customer Dimensi..	CRM, ECOM	mc_analytics.custo..	Hourly	00:10-23:50	+15 min	100% CAE SRED-..
ETL.CUST.FACT...	Customer Fact Loa..	Events, Ledger	mc_analytics.custo..	Hourly	00:20-23:59	+20 min	99.2% CAE SRED..
ETL.CUST.SEG.E..	Segmentation Enri..	LKP, Models	mc_analytics.custo..	2-hourly	01:00-23:00	+30 min	100% Customer A..
ETL.CUST.DQ.CH..	EDQ Quality Chec..	Dim/Fact	mc_dataops.edq_a..	Hourly	00:30-23:30	+25 min	100% EDQ EDQ-R..
ETL.CUST.LKP.R..	Lookup Refresh	Ref Data	mc_refdata.*	Daily	02:00-02:20	02:30	100% Data Platfor..
ETL.CUST.HOUS..	Household Linkage	CRM	mc_analytics.custo..	4-hourly	01:30-21:30	+40 min	100% CAE SRED-..
ETL.CUST.RECO..	Ledger Reconciliati..	Ledger	mc_analytics.reco..	Daily	03:00-03:40	04:00	100% Finance Dat..
ETL.CUST.EXPO..	Marketing Export	Dim/Fact	mkt_exports.custo..	Daily	04:00-04:30	05:00	100% Marketing A..
ETL.CUST.GEO.E..	Geocoding Enrich..	Vendors	mc_analytics.custo..	Daily	01:10-01:40	02:00	100% Location Int..
ETL.CUST.AGG.M..	Monthly Aggregati..	Fact	mc_analytics.custo..	Daily	00:40-01:05	01:20	100% CAE SRED-..
ETL.CUST.BACKF..	Controlled Backfills	Various	mc_analytics.*	On-Demand	As Scheduled	Per Runbook	N/A CAE SRED-C..
ETL.CUST.DATAF..	Vendor Feed Integ..	Vendor X	mc_external.vendo..	Daily	22:00-22:20	22:45	100% Data Platfor..

Throughput and Volume Metrics -- ETL.CUST.FACT.LOAD.v5 (Trailing 14 Days)

Date (UTC)	Batches	Rows Extracted	Rows Loaded	Avg Throughput (..	Peak Memory (GB)	Runtime (hh:mm:..	Status Notes
2026-07-05	24	12,481,902	12,481,902	14,220	6.1	00:14:51	Success Nominal l..
2026-07-06	24	12,709,144	12,709,144	14,305	6.0	00:14:48	Success Nominal l..
2026-07-07	24	12,512,877	12,512,877	14,198	6.1	00:14:41	Success Nominal l..
2026-07-08	24	12,733,420	12,733,420	14,276	6.1	00:14:51	Success Slight incr..
2026-07-09	24	12,801,003	12,801,003	14,281	6.2	00:14:57	Success Nominal
2026-07-10	24	12,766,918	12,766,918	14,289	6.2	00:14:53	Success Nominal
2026-07-11	24	12,550,104	12,550,104	14,207	6.1	00:14:45	Success Weekend..
2026-07-12	24	12,498,662	12,498,662	14,185	6.1	00:14:42	Success Weekend
2026-07-13	24	12,905,771	12,905,771	14,332	6.2	00:14:59	Success Nominal
2026-07-14	24	12,988,330	12,988,330	14,361	6.2	00:15:03	Success Minor upli..
2026-07-15	24	12,941,552	12,941,552	14,348	6.2	00:15:00	Success Nominal
2026-07-16	24	12,876,441	12,876,341	14,112	6.4	00:15:13	Success (Warn) 10..
2026-07-17	24	12,922,884	12,922,884	14,336	6.2	00:14:58	Success Normaliz..
2026-07-18	24	12,911,220	12,911,220	14,329	6.2	00:14:57	Success End of wi..

Performance Snapshot -- Additional Jobs (2026-07-12 to 2026-07-18)

Job ID	Min Runtime (mm..	Max Runtime (m..	Avg Runtime (m..	95p Runtime (mm..	Avg Rows	Success Rate	Last Failure Capa..
ETL.CUST.DIM.L..	04:51	06:12	05:28	06:03	2,104,992	100%	N/A 38% Stable; C..
ETL.CUST.SEG.E..	07:11	09:22	08:04	09:01	2,100,331	100%	N/A 41% Model ref..
ETL.CUST.DQ.CH..	03:42	04:19	03:59	04:14	14,000 checks	100%	N/A 47% Rule exp..
ETL.CUST.LKP.R..	06:33	07:05	06:51	07:01	120,440	100%	N/A 52% Vendor f..
ETL.CUST.HOUS..	05:59	07:14	06:21	07:02	820,144	100%	N/A 44% No drift
ETL.CUST.RECO..	18:42	22:14	20:03	21:51	N/A	100%	N/A 29% Variance ..
ETL.CUST.EXPO..	08:10	09:36	08:52	09:29	3,411,992	100%	N/A 36% Downstre..
ETL.CUST.GEO.E..	10:15	11:44	10:58	11:29	410,222	100%	N/A 33% Batch ge..

ETL.CUST.AGG.M..	12:01	13:12	12:31	13:01	12,911,220	100%	N/A 35% Daily rec..
ETL.CUST.DATAF..	04:40	05:19	05:01	05:16	98,221	100%	N/A 49% Ingest lat..

Exception Summary -- 2026-07-12 to 2026-07-18

Exception Type	Count	Impacted Job(s)	Severity	Reference	Resolution State	Notes
Late-arriving duplic..	1	ETL.CUST.FACT...	Low	INC-58674	Closed 2026-07-1..	100 rows quaranti..
Upstream lookup d..	0	ETL.CUST.SEG.E..	N/A	N/A	N/A	No events
Volume spike >5%	1	ETL.CUST.FACT...	Low	MRC-ETL-4302	Monitoring	Aligned to promo ..
Intermittent vendor..	0	ETL.CUST.DATAF..	N/A	N/A	N/A	No events
Reconciliation vari..	0	ETL.CUST.RECO..	N/A	N/A	N/A	Within tolerance

Operational Notes:

- SLA adherence across Customer Analytics v5 maintained at 99.9% for the period; no priority-1 incidents recorded. The single warning on 2026-07-16 was addressed with automated quarantine and post-load reconciliation per EDQ standard operating controls.
- Capacity headroom remains above 29% for all listed jobs; SRE-Data will continue to track utilization via dashboard DS-CUST-ETL-Perf v3.2. Next capacity review is scheduled under CAB CHG-2026-0724 (2026-07-24 16:00 UTC) with CAE, EDQ, and Finance Data Ops stakeholders.
- Any schedule or SLA adjustments will be raised via standard change management with minimum 48-hour notice and referenced in subsequent publication of this specification (v5.x minor updates).

Page 11 -- Schema Change Register and Lineage Validation (v5 Stabilization Window)

This page records the approved, routine field-level adjustments and lineage confirmations performed under ETL Schema Specification -- Customer Analytics v5 for Meridian Capital. The changes below were reviewed by Data Engineering, Data Governance, and Analytics Enablement during the Q3 FY26 cutover window. All entries reference the governing tickets, timestamps, and accountable teams. Unless otherwise stated, adjustments are backward compatible for consuming analytics views and reporting marts. Lineage verifications focus on target-to-source traceability for critical facts and dimensions across CRM, Core Banking, and Digital channels and are aligned with the enterprise data catalog entries. No policy exceptions were requested or granted; all items followed the standard CAB process.

Schema Change Register (Approved -- Q3 FY26)

Object (Target)	Field	Change Type	Rationale	Downstreams Im..	Backfill/Default St..	Compatibility	Ticket Ref Review..
dim_customer	email_hash	New Field	Enable privacy-pre..	7	Derived from lower..	Backward	MCA-ETL-7324 J. ..
dim_customer	loyalty_tier	Type Widen (VAR..	Standardize tier la..	5	Copy existing valu..	Backward	MC-ANL-8812 S. ..
fact_interactions_d..	session_duration_..	Constraint Update ..	Enforce non-negati..	4	Nullify negatives; r..	Backward	DQ-5641 R. Nguye..
fact_interactions_d..	channel_code	Value Domain Upd..	Align codes with Di..	9	Map legacy codes ..	Backward	MCA-ETL-7359 L. ..
dim_product	category_path	New Field	Support hierarchic..	3	Build from product ..	Backward	MC-ANL-8791 E. ..
dim_region	market_code	Comment Update	Clarify ISO-3166 al..	2	N/A	N/A	DOC-1129 P. Stei..
vw_customer_profi..	preferred_lang	Deprecate (soft)	Superseded by co..	6	Keep populated; fl..	Backward	MCA-ETL-7402 T. ..
dim_customer	comms_language	New Field	Unambiguous com..	6	Default to 'en' if mi..	Backward	MC-ANL-8827 M. ..
stg_crm_customer	phone_clean	Type Widen (VAR..	Support E.164 with..	1	Recompute on ing..	Backward	MCA-ETL-7444 F. ..
fact_customer_daily	last_purchase_ts	Type Widen (NTZ-..	Preserve timezone..	8	Convert NTZ to UT..	Backward	MC-ANL-8843 H. ..
fact_customer_daily	active_flag	Default Update	Ensure explicit boo..	4	Default to true for ..	Backward	DQ-5673 R. Lopez..
dim_customer	birth_date	Constraint Update ..	Filter spurious dates	3	Null-out invalid dat..	Backward	DQ-5682 C. Allen (..
fact_transactions	merchant_categor..	New Field	Support spend seg..	10	Backfill from card n..	Backward	MCA-ETL-7481 B. ..
dim_channel	channel_group	Value Domain Upd..	Harmonize with Int..	3	Lookup from mapp..	Backward	MC-ANL-8869 D. ..
vw_customer_profi..	tenure_days	Formula Update	Switch to business..	5	Recompute nightly	Backward	MCA-ETL-7512 K. ..
dim_product	sku	Uniqueness Constr..	Enforce SKU uniqu..	2	Resolve dupes via ..	Backward	DQ-5718 V. Ivanov..
fact_interactions_d..	engagement_score	New Field	Introduce normaliz..	6	Backfill last 90 days	Backward	MC-ANL-8892 Y. ..
dim_customer	gdpr_consent_flag	Comment Update	Expand definition t..	4	N/A	N/A	DOC-1142 J. Patel..
stg_digital_event	event_id	Type Widen (INT->..	Growth in event vo..	1	Cast on load	Backward	MCA-ETL-7547 F. ..
fact_transactions	fx_rate_applied	New Field	Transparent FX co..	5	Backfill from treasu..	Backward	MC-ANL-8904 S. ..
dim_region	region_id	Surrogate Key Seq..	Exhaustion thresh..	2	Advance sequence..	Backward	OPS-CHG-4207 P...
vw_customer_profi..	risk_segment	Value Domain Upd..	Align with Enterpri..	4	Map old tiers to new	Backward	MC-ANL-8920 L. ..
fact_customer_daily	avg_balance_30d	Formula Update	Move to median fo..	3	Recompute rolling ..	Backward	MCA-ETL-7589 H. ..
dim_product	is_discontinued	New Field	Support inventory ..	2	Default false	Backward	MC-ANL-8936 E. ..
fact_transactions	chargeback_flag	New Field	Improve dispute an..	7	Default false; backf..	Backward	MCA-ETL-7605 B. ..
dim_customer	household_id	New Field	Household-level a..	6	Derive from addres..	Backward	MC-ANL-8951 M. ..

Field Lineage Map (Excerpts -- Post-Change v5)

Target Object.Fiel..	Source System	Source Object.Fi..	Transform/Rule S..	Data Steward	Validation Ticket	Last Validated (U..
dim_customer.em..	CRM	crm_contact.email	Lowercase; hash t..	J. Patel (DG)	DQ-5665	2026-06-18T11:09..
dim_customer.com..	CRM	crm_contact.lang_..	Normalize to ISO 6..	M. Costa (DG)	MC-ANL-8827	2026-06-18T10:58..
fact_customer_dail..	Core Banking	txn_header.posted..	Convert to UTC; re..	H. Park (DE)	MC-ANL-8843	2026-06-19T10:21..
fact_interactions_d..	Digital	web_event.channe..	Map via taxonomy..	L. Ortiz (AE)	MCA-ETL-7359	2026-06-16T09:44..
dim_product.categ..	PIM	pim_category.path	Concatenate hiera..	E. Kim (DE)	MC-ANL-8791	2026-06-16T16:03..
stg_crm_customer..	CRM	crm_contact.phon..	Standardize to E.1..	F. Singh (DE)	MCA-ETL-7444	2026-06-18T17:02..
fact_transactions...	Card Network	auth.mcc	Left-pad to 4 chars..	S. Wu (DE)	MCA-ETL-7481	2026-06-20T13:07..
dim_region.market..	Reference	iso_region.alpha2	Direct reference; n..	P. Stein (DG)	DOC-1129	2026-06-17T07:21..
vw_customer_profi..	CRM + Core Banki..	acct.open_dt, sysd..	Business days bet..	K. Adams (AE)	MCA-ETL-7512	2026-06-21T17:11..
dim_product.sku	PIM	pim_product.sku_r..	Trim; uppercase; u..	V. Ivanov (QAV)	DQ-5718	2026-06-22T09:40..

fact_interactions_d..	Digital	event weights, ses..	Weighted composi..	Y. Chen (DE)	MC-ANL-8892	2026-06-22T13:22..
dim_customer.birt..	CRM	crm_contact.dob	Null if out of range ..	C. Allen (DG)	DQ-5682	2026-06-20T08:33..
fact_transactions.f..	Treasury	fx_daily.rates	Join on posting dat..	S. Wu (DE)	MC-ANL-8904	2026-06-24T10:32..
dim_customer.hou..	CRM + Address	crm_contact.addr_..	Stable key: hashe..	M. Costa (DG)	MC-ANL-8951	2026-06-27T10:51..
vw_customer_profi..	Risk	risk_customer.seg..	Map v1->v2 via ref..	L. Ortiz (AE)	MC-ANL-8920	2026-06-25T09:58..
fact_customer_dail..	Core Banking	balance_snap.dail..	30-day rolling medi..	H. Park (DE)	MCA-ETL-7589	2026-06-25T13:44..
dim_product.is_dis..	PIM	pim_product.lifecy..	True if status in {'r..	E. Kim (DE)	MC-ANL-8936	2026-06-26T08:22..
fact_transactions.c..	Disputes	case_header.status	True if case opene..	B. Shah (AE)	MCA-ETL-7605	2026-06-26T12:41..
fact_interactions_d..	Digital	session.end_ts, se..	Max(end - start, 0) ..	R. Nguyen (QAV)	DQ-5641	2026-06-15T14:02..

Page 12 -- Data Quality Outcomes and Operational Metrics (June FY26)

This page summarizes data quality checks and operational performance for the Customer Analytics v5 pipeline in June FY26. The quality measures emphasize completeness, conformance to domains, uniqueness of business keys, referential integrity across facts and dimensions, and timeliness against ingestion and publish SLAs. Operational metrics aggregate the June production runs across core DAGs. All measures were captured by the QA & Validation schedule and reviewed weekly with Platform Operations. Minor variances were tracked via standard incident tickets and closed with routine remediations. No material data loss or SLA breach impacted downstream consumers during the period.

Data Quality Checks -- Monthly Summary (June FY26)

Check ID	Target Object.Field	Dimension	Rule Summary	Threshold	June Result	Status	Owner Team Tick..
DQ-CPL-0001	dim_customer.cust..	Uniqueness	No duplicate surro..	100% unique	100%	Pass	QAV RUN-202606..
DQ-CPL-0002	dim_customer.ema..	Completeness	Null rate <= 15%	<=15% null	12.8%	Pass	QAV RUN-202606..
DQ-CPL-0003	dim_customer.birth..	Validity	Date >= 1900-01-01	>=99.5% valid	99.8%	Pass	DG DQ-5682 2026..
DQ-CPL-0004	fact_transactions.t..	Uniqueness	Unique per posting..	100% unique	100%	Pass	QAV RUN-202606..
DQ-CPL-0005	fact_transactions.a..	Validity	Amount >= 0 exce..	>=99.9% valid	99.93%	Pass	DE RUN-2026062..
DQ-CPL-0006	fact_transactions.c..	Domain	In ISO-4217 3-lette..	>=99.9% valid	100%	Pass	DG RUN-2026062..
DQ-CPL-0007	fact_customer_dail..	Timeliness	Max lag <= 24h	<=24h	6h	Pass	AE MC-ANL-8843 ..
DQ-CPL-0008	dim_product.sku	Uniqueness	SKU unique acros..	100% unique	100%	Pass	QAV DQ-5718 202..
DQ-CPL-0009	dim_region.market..	Domain	ISO alpha-2 region..	>=99.9% valid	100%	Pass	DG RUN-2026062..
DQ-CPL-0010	fact_interactions_d..	Domain	In taxonomy v3 co..	>=99.0% valid	99.6%	Pass	AE MCA-ETL-735..
DQ-CPL-0011	fact_interactions_d..	Validity	Duration >= 0	>=99.95% valid	99.98%	Pass	QAV DQ-5641 202..
DQ-CPL-0012	fact_transactions.f..	Completeness	Populated when cu..	>=99.5% filled	99.7%	Pass	DE MC-ANL-8904 ..
DQ-CPL-0013	vw_customer_profi..	Accuracy	Business days calc..	+/-1 day	Within tol	Pass	AE MCA-ETL-751..
DQ-CPL-0014	dim_customer.com..	Domain	ISO 639-1 two-lett..	>=99.5% valid	99.9%	Pass	DG MC-ANL-8827 ..
DQ-CPL-0015	dim_product.is_dis..	Validity	Boolean in {true,fal..	100% valid	100%	Pass	DE MC-ANL-8936 ..
DQ-CPL-0016	fact_transactions.c..	Consistency	True implies disput..	100% consistent	100%	Pass	AE MCA-ETL-760..
DQ-CPL-0017	fact_customer_dail..	Accuracy	Median aligns with ..	+/-0.5%	0.2% diff	Pass	DE MCA-ETL-758..
DQ-CPL-0018	dim_customer.hou..	Stability	Key stable across r..	>=99.99% stable	100%	Pass	DG MC-ANL-8951 ..
DQ-CPL-0019	fact_interactions_d..	Range	In [0,1] inclusive	100% in range	100%	Pass	QAV RUN-202606..
DQ-CPL-0020	stg_digital_event.e..	Capacity	No overflow after t..	0 overflow	0	Pass	DE MCA-ETL-754..
DQ-CPL-0021	vw_customer_profi..	Domain	In Risk v2 tiers	>=99.5% valid	99.7%	Pass	AE MC-ANL-8920 ..
DQ-CPL-0022	fact_transactions.p..	Timeliness	Posted <= ingest_..	>=99.0% on time	99.4%	Pass	QAV RUN-202606..
DQ-CPL-0023	dim_customer.gdp..	Consistency	True requires cons..	100% consistent	100%	Pass	DG DOC-1142 202..
DQ-CPL-0024	dim_channel.chan..	Domain	Matches Interactio..	>=99.0% valid	99.2%	Pass	DE MC-ANL-8869 ..
DQ-CPL-0025	fact_customer_dail..	Completeness	Default applied on ..	>=99.9% filled	100%	Pass	QAV DQ-5673 202..
DQ-CPL-0026	dim_product.categ..	Completeness	Non-null for classifi..	>=98.0% filled	98.7%	Pass	DE MC-ANL-8791 ..
DQ-CPL-0027	fact_transactions...	Validity	Numeric; 4 chars p..	>=99.9% valid	99.95%	Pass	DE MCA-ETL-748..
DQ-CPL-0028	dim_region.region..	Referential	All facts reference ..	100% enforced	100%	Pass	DG OPS-CHG-420..
DQ-CPL-0029	fact_interactions_d..	Uniqueness	Unique per day per..	>=99.9% unique	99.92%	Pass	QAV RUN-202606..
DQ-CPL-0030	vw_customer_profi..	Monitoring	Deprecated field st..	Informational	100%	Info	AE MCA-ETL-740..

Operational Metrics -- ETL Jobs (June FY26, PROD)

Job/DAG	Environment	Avg Start (UTC)	Avg Duration	Avg Rows In	Avg Rows Out	Reject Rate	SLA Target SLA ..
DAG_CRM_DIM_..	PROD	02:05	00:24	4,210,553	4,210,553	0.03%	00:45 100 0 2026-..
DAG_DIGITAL_FA..	PROD	02:20	00:37	52,114,782	51,982,441	0.25%	01:00 100 1 2026-..
DAG_COREBANK..	PROD	02:40	00:31	9,842,116	9,842,116	0.01%	01:00 100 0 2026-..
DAG_CARDNET_..	PROD	03:05	00:43	18,229,441	18,225,337	0.02%	01:15 100 0 2026-..
DAG_PIM_DIM_P..	PROD	01:55	00:18	512,903	512,903	0.00%	00:30 100 0 2026-..
DAG_REF_REGIO..	PROD	01:35	00:07	289	289	0.00%	00:20 100 0 2026-..
DAG_PROFILE_VI..	PROD	04:05	00:19	7,311,020	7,311,020	0.00%	00:30 100 0 2026-..

DAG_QA_VALIDA..	PROD	04:30	00:12	90 checks	90 results	N/A	00:20 100 0 2026-..
DAG_TREASURY..	PROD	03:35	00:09	143	143	0.00%	00:20 100 0 2026-..
DAG_DISPUTES_..	PROD	03:50	00:08	5,441	5,441	0.00%	00:15 100 0 2026-..
DAG_AE_TAXON..	PROD	01:15	00:05	87	87	0.00%	00:15 100 0 2026-..
DAG_DE_DUPLIC..	PROD	02:00	00:06	1,234	1,232	0.16%	00:20 100 0 2026-..
DAG_AUDIT_BAL..	PROD	03:15	00:10	9,842,116	9,842,116	0.00%	00:20 100 0 2026-..
DAG_RISK_SEG..	PROD	03:25	00:07	7,311,020	7,311,020	0.00%	00:20 100 0 2026-..
DAG_HOUSEHOL..	PROD	02:50	00:09	4,210,553	4,210,553	0.00%	00:20 100 0 2026-..

June Operational Notes -- Incidents and Follow-ups (Informational)

Incident ID	Severity	Date (UTC)	Affected Job	Symptom	Root Cause	Resolution	Owner Closed (U..
OPS-INC-6741	Low	2026-06-12T02:57:..	DAG_DIGITAL_FA..	Slight throughput d..	Upstream batch si..	Auto-scaled worke..	Platform Ops 2026..
OPS-INC-6732	Info	2026-06-05T02:31:..	DAG_CRM_DIM_..	Retry observed	Transient connecto..	Standard retry suc..	Data Engineering ..
OPS-INC-6738	Info	2026-06-08T03:13:..	DAG_COREBANK..	Metric spike warni..	Month-end volume..	Threshold adjusted..	QAV 2026-06-08T..
OPS-INC-6746	Info	2026-06-18T03:49:..	DAG_CARDNET_..	Delay warning	Card network late f..	Consumed on arriv..	Platform Ops 2026..
OPS-INC-6750	Info	2026-06-22T04:24:..	DAG_PROFILE_VI..	Cache refresh note	Stale cache on BI l..	Cache invalidated ..	Analytics Enablem..

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Page 13: Field Conformance Review Summary

This section captures the cross-object field conformance review conducted by Customer Analytics ETL and Data Governance during the v5.0.4 validation window. The review covered production runs between 2026-07-15 and 2026-07-21 (UTC) and reconciled observed structures and nullability against the specification baseline approved under CAB record CAB-CA-2026-0711. Findings were logged in the MCA Customer Analytics board with references below. No blocking defects were identified; advisory items are tracked for remediation in the next minor increment.

Field Conformance Matrix (extract window: 2026-07-21T03:00:00Z-2026-07-22T03:00:00Z; baseline spec tag: ca-v5.0.4)

Object	Field	Data Type (Spec)	Data Type (Obser..	Nullable (Spec)	Nullable (Observe..	Default (Spec)	Rule ID Validation..
dim_customer	customer_id	STRING	STRING	No	No	n/a	DQ-CA-001 Pass ..
dim_customer	external_ref_id	STRING	STRING	Yes	Yes	null	DQ-CA-002 Pass ..
dim_customer	birth_date	DATE	DATE	Yes	Yes	null	DQ-CA-003 Pass ..
dim_customer	email_hash	STRING	STRING	Yes	Yes	null	DQ-CA-004 Pass ..
dim_customer	is_opted_in	BOOLEAN	BOOLEAN	No	No	false	DQ-CA-005 Pass ..
dim_customer	create_ts	TIMESTAMP	TIMESTAMP	No	No	current_timestamp	DQ-CA-006 Pass ..
dim_customer	update_ts	TIMESTAMP	TIMESTAMP	No	Yes	current_timestamp	DQ-CA-007 Warni..
dim_account	account_id	STRING	STRING	No	No	n/a	DQ-CA-011 Pass ..
dim_account	customer_id	STRING	STRING	No	No	n/a	DQ-CA-012 Pass ..
dim_account	open_date	DATE	DATE	No	No	n/a	DQ-CA-013 Pass ..
dim_account	status_cd	STRING	STRING	No	No	'ACTIVE'	DQ-CA-014 Pass ..
dim_account	segment_cd	STRING	STRING	Yes	Yes	null	DQ-CA-015 Pass ..
dim_channel	channel_id	STRING	STRING	No	No	n/a	DQ-CA-021 Pass ..
dim_channel	channel_nm	STRING	STRING	No	No	n/a	DQ-CA-022 Pass ..
dim_channel	active_flg	BOOLEAN	BOOLEAN	No	No	true	DQ-CA-023 Pass ..
dim_region	region_id	STRING	STRING	No	No	n/a	DQ-CA-031 Pass ..
dim_region	region_nm	STRING	STRING	No	No	n/a	DQ-CA-032 Pass ..
dim_region	parent_region_id	STRING	STRING	Yes	Yes	null	DQ-CA-033 Pass ..
dim_time	date_key	DATE	DATE	No	No	n/a	DQ-CA-041 Pass ..
dim_time	week_of_year	INT64	INT64	No	No	n/a	DQ-CA-042 Pass ..
dim_time	fiscal_qtr	STRING	STRING	No	No	n/a	DQ-CA-043 Pass ..
fact_interaction	interaction_id	STRING	STRING	No	No	n/a	DQ-CA-051 Pass ..
fact_interaction	customer_id	STRING	STRING	No	No	n/a	DQ-CA-052 Pass ..
fact_interaction	channel_id	STRING	STRING	No	No	n/a	DQ-CA-053 Pass ..
fact_interaction	occurred_ts	TIMESTAMP	TIMESTAMP	No	No	n/a	DQ-CA-054 Pass ..
fact_interaction	interaction_type_cd	STRING	STRING	No	No	n/a	DQ-CA-055 Pass ..
fact_interaction	duration_seconds	INT64	INT64	Yes	Yes	null	DQ-CA-056 Pass ..
fact_transaction	txn_id	STRING	STRING	No	No	n/a	DQ-CA-061 Pass ..
fact_transaction	account_id	STRING	STRING	No	No	n/a	DQ-CA-062 Pass ..
fact_transaction	txn_amt	DECIMAL(18,2)	DECIMAL(18,2)	No	No	0.00	DQ-CA-063 Pass ..
fact_transaction	txn_ts	TIMESTAMP	TIMESTAMP	No	No	n/a	DQ-CA-064 Pass ..
fact_transaction	currency_cd	STRING	STRING	No	No	'USD'	DQ-CA-065 Warni..
fact_transaction	merchant_categor..	STRING	STRING	Yes	Yes	null	DQ-CA-066 Pass ..
bridge_customer_s..	customer_id	STRING	STRING	No	No	n/a	DQ-CA-071 Pass ..
bridge_customer_s..	segment_cd	STRING	STRING	No	No	n/a	DQ-CA-072 Pass ..
bridge_customer_s..	as_of_date	DATE	DATE	No	No	n/a	DQ-CA-073 Pass ..
dim_product	product_id	STRING	STRING	No	No	n/a	DQ-CA-081 Pass ..
dim_product	product_nm	STRING	STRING	No	No	n/a	DQ-CA-082 Pass ..
dim_product	product_family_cd	STRING	STRING	Yes	Yes	null	DQ-CA-083 Pass ..
dim_product	active_flg	BOOLEAN	BOOLEAN	No	No	true	DQ-CA-084 Pass ..

Semantic Domain Alignment Observations (categorical coverage checks; sample window aligned to above)

Domain	Field	Reference Set (S..	Coverage Observ..	Out-of-Set Values..	Status	Ticket Ref	Steward
Country Codes	dim_customer.cou..	ISO-3166-1 alpha-2	100%	0	Pass	n/a	Data Governance
Currency Codes	fact_transaction.cu..	ISO-4217	99.4%	11	Warning	MCA-CA-7431	Data Governance
Segment Codes	dim_account.segm..	SEGMENT_V5 co..	100%	0	Pass	n/a	Customer Analytic..
Interaction Types	fact_interaction.int..	INTXN_TYPE_V5	99.2%	5	Warning	MCA-CA-7446	Data Governance
Merchant Category	fact_transaction.m..	MCC_V2026_Q3	98.7%	19	Warning	MCA-CA-7444	Customer Analytic..
Channel	fact_interaction.ch..	CHANNEL_DIM	100%	0	Pass	n/a	Data Governance
Product Family	dim_product.produ..	PROD_FAMILY_V5	99.8%	2	Pass	MCA-CA-7440	Data Governance
Region	dim_region.region..	REGION_DIM	100%	0	Pass	n/a	Customer Analytic..

Notes:

- Nullable relaxation observed on dim_customer.update_ts is under review; no downstream impact reported. Advisory retained to realign loader behavior with baseline defaulting semantics (MCA-CA-7441).
- Currency code variance traced to late-arriving non-USD markets during backfill. Governance approved temporary exception pending full code list refresh in v5.0.5 (MCA-CA-7431).

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Page 14: Pipeline Metrics and SLA Adherence

The following operational metrics summarize ETL job performance and delivery timeliness for Customer Analytics v5 across Production during the week ending 2026-07-21 (UTC). Measurements are aggregated by job for the last seven scheduled runs. SLA thresholds reference OPS-SLA-CA-2026-01. No customer-facing data latency breaches occurred; two internal advisories are carried forward to the next sprint.

ETL Job Runtime and Throughput (Production; window: 2026-07-15-2026-07-21 UTC)

Job ID	Job Name	Schedule Windo..	Runs (7d)	Success Rate	P95 Duration (m..	Mean Duration (m..	Rows In (avg) Ro..
CA-EXT-001	Extract: Customer ..	02:00-02:45	7	100%	22:14	20:58	1,245,330 1,245,3..
CA-EXT-002	Extract: Account M..	02:15-03:00	7	100%	24:05	22:37	885,412 885,412 0..
CA-EXT-003	Extract: Product C..	01:30-02:00	7	100%	09:18	08:56	12,404 12,404 0 0..
CA-EXT-004	Extract: Channel R..	01:10-01:25	7	100%	03:51	03:45	128 128 0 00:12 2..
CA-TRF-101	Transform: dim_cu..	03:00-03:30	7	100%	12:33	11:47	1,245,330 1,245,3..
CA-TRF-102	Transform: dim_ac..	03:10-03:35	7	100%	10:42	10:19	885,412 885,412 0..
CA-TRF-103	Transform: dim_ch..	02:50-03:05	7	100%	03:02	02:57	128 128 0 00:57 2..
CA-TRF-104	Transform: dim_re..	02:55-03:10	7	100%	03:49	03:41	42 42 0 00:59 202..
CA-TRF-105	Transform: dim_ti..	02:00-02:10	7	100%	01:08	01:07	365 365 0 00:09 2..
CA-TRF-201	Transform: fact_int..	03:30-04:10	7	100%	28:55	27:42	6,744,221 6,742,1..
CA-TRF-202	Transform: fact_tra..	03:35-04:20	7	100%	31:12	29:48	4,209,993 4,209,9..
CA-ENR-301	Enrich: Customer ..	04:15-04:35	7	100%	09:44	09:21	1,245,330 1,245,3..
CA-ENR-302	Enrich: Merchant ..	04:20-04:45	7	100%	08:57	08:39	4,209,993 4,209,9..
CA-ENR-303	Enrich: Currency N..	04:25-04:50	7	100%	06:12	06:01	4,209,993 4,209,9..
CA-LD-401	Load: dim_customer	04:30-04:50	7	100%	07:18	07:04	1,245,330 1,245,3..
CA-LD-402	Load: dim_account	04:35-04:55	7	100%	06:47	06:33	885,412 885,412 0..
CA-LD-403	Load: dimensions (..	04:10-04:25	7	100%	04:22	04:15	535 535 0 02:03 2..
CA-LD-404	Load: fact_interacti..	04:45-05:25	7	100%	34:11	33:02	6,742,106 6,742,1..
CA-LD-405	Load: fact_transact..	04:50-05:35	7	100%	35:48	34:09	4,209,993 4,209,9..
CA-VAL-501	Validate: Referenti..	05:30-05:45	7	100%	05:19	05:08	n/a n/a n/a 03:24 2..
CA-VAL-502	Validate: Codelist ..	05:30-05:50	7	100%	07:26	07:12	n/a n/a n/a 03:26 2..
CA-AGG-601	Publish: Analytics ..	06:00-06:20	7	100%	12:42	12:02	12,287,962 12,287..
CA-AGG-602	Publish: Interaction..	06:05-06:25	7	100%	10:51	10:26	6,742,106 6,742,1..
CA-AGG-603	Publish: Revenue ..	06:10-06:30	7	100%	11:09	10:47	4,209,993 4,209,9..

SLA Exceptions and Advisory Items (week ending 2026-07-21; thresholds per OPS-SLA-CA-2026-01)

Date (UTC)	Job ID	SLA Threshold	Actual	Breach Type	Impact Summary	Status	Ticket Ref Owner ..
2026-07-16	CA-TRF-201	Error Rows <= 0.0..	0.03%	Near-threshold	Transient spikes in..	Monitoring	MCA-CA-7442 Cu..
2026-07-17	CA-ENR-303	Codelist pass >= 9..	99.4%	Advisory	Currency code set ..	In Progress	MCA-CA-7431 Dat..
2026-07-18	CA-TRF-102	Duration P95 <= 1..	10:42	Compliant	No action required	Closed	n/a Customer Anal..
2026-07-19	CA-LD-405	Freshness <= 03:15	03:03	Compliant	No action required	Closed	n/a Data Platform ..
2026-07-19	CA-VAL-502	Coverage >= 99.5%	99.4%	Advisory	Tied to currency n..	In Progress	MCA-CA-7431 QA ..
2026-07-20	CA-TRF-201	Error Rows <= 0.0..	0.03%	Near-threshold	Continued small p..	Monitoring	MCA-CA-7442 Cu..
2026-07-20	CA-EXT-002	Start by 02:20	02:17	Compliant	Upstream feed arri..	Closed	n/a Data Platform ..
2026-07-21	CA-ENR-302	Enrichment P95 <..	08:57	Compliant	Merchant map enri..	Closed	n/a Data Governan..

Operational Notes:

- Interaction transform (CA-TRF-201) maintained stability with near-threshold error ratios linked to optional interaction duration fields. Observed pattern remains within approved analytic tolerance; continued observation through Sprint 2026.29 (MCA-CA-7442).

- Currency normalization advisories (CA-ENR-303, CA-VAL-502) are correlated to scheduled code list updates. No downstream KPI variance beyond rounding tolerances. Target resolution is aligned to codelist release window in v5.0.5 (MCA-CA-7431).
- Overall daily mart publication times held steady with P95 12:42 and freshness lag under four hours, meeting reporting cadence for executive dashboards.

End of Pages 13-14.

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Page 15 -- Section 8: Field-Level Conformance Results (Q2 FY26 Review)

As part of the Q2 FY26 schema conformance review, the Customer Analytics QA guild validated field-level data quality against the v5 specification across customer_profile, customer_activity, and marketing_attribution entities. Sampling used 10% stratified draws across prime source systems (CRM-Core, Ledger-360, OutreachHub) between 2026-04-01 and 2026-06-30. Thresholds were applied per Data Governance controls (DQ-Policy-12.3). Findings were logged via ServiceNow/JIRA hybrids and adjudicated in triage sessions with Data Engineering, Customer Analytics, and Domain Stewards. The table below enumerates observed deviations, null rate movements, distinct cardinality shifts, and out-of-domain instances with remediation dispositions. All timestamps are UTC; ownership reflects accountable teams for follow-up.

Field-Level Conformance Findings

Field Name	Source System	Expected Type/F..	Observed Variants	Null Rate (Prev)	Null Rate (Curr)	Distinct Count De..	Out-of-Domain In..
customer_id	CRM-Core	STRING(36) UUID	None	0.00%	0.00%	+0.4%	0 MC-DQ-10421 C..
account_id	Ledger-360	STRING(24)	None	0.12%	0.11%	+1.2%	0 MC-DQ-10422 M..
email_address	OutreachHub	STRING RFC5322	Missing TLD, mixe..	1.86%	1.92%	+2.8%	147 MC-DQ-10423..
phone_number_e1..	CRM-Core	STRING E.164	Local formats	0.95%	0.88%	+1.5%	62 MC-DQ-10424 ..
birth_date	CRM-Core	DATE (YYYY-MM-..	0000-01-01 sentin..	3.21%	3.08%	+0.9%	24 MC-DQ-10425 ..
gender_code	CRM-Core	ENUM [M,F,O,U]	Lowercase variants	0.76%	0.74%	+0.3%	11 MC-DQ-10426 ..
address_line1	CRM-Core	STRING(255)	Truncation risk	0.58%	0.57%	+0.6%	0 MC-DQ-10427 S..
address_line2	CRM-Core	STRING(255)	None	18.7%	18.9%	+1.1%	0 MC-DQ-10428 A..
city	CRM-Core	STRING(120)	Accented chars	0.63%	0.61%	+0.7%	0 MC-DQ-10429 C..
state_province	CRM-Core	STRING(64)	Abbrev/long mix	0.69%	0.64%	-1.8%	19 MC-DQ-10430 ..
postal_code	CRM-Core	STRING(16)	Spaces/dashes	1.14%	1.10%	+1.1%	36 MC-DQ-10431 ..
country_code	CRM-Core	STRING ISO-3166..	alpha-3 input	0.07%	0.06%	+0.2%	9 MC-DQ-10432 M..
segment_code	Analytics	ENUM [A,B,C,D]	Legacy 'E' seen	0.00%	0.00%	+0.0%	5 MC-DQ-10433 P..
risk_score	RiskHub	DECIMAL(5,2) 0..1..	>100 edge values	0.09%	0.09%	+0.5%	3 MC-DQ-10434 C..
lifetime_value_usd	Ledger-360	DECIMAL(18,2)	Negatives	0.43%	0.39%	-0.8%	27 MC-DQ-10435 ..
first_purchase_dt	Ledger-360	DATE	1970-01-01 sentinel	2.41%	2.36%	-0.6%	14 MC-DQ-10436 ..
last_activity_ts	Events-EDL	TIMESTAMP(UTC)	TZ offsets	0.37%	0.35%	+0.4%	18 MC-DQ-10437 ..
opt_in_email	OutreachHub	BOOLEAN	'Y'/'N' strings	0.00%	0.00%	+0.0%	41 MC-DQ-10438 ..
opt_in_sms	OutreachHub	BOOLEAN	'true' lower-case	0.00%	0.00%	+0.0%	23 MC-DQ-10439 ..
marketing_source	OutreachHub	STRING(64)	Mixed casing	4.12%	4.05%	-0.9%	0 MC-DQ-10440 L..
campaign_id	OutreachHub	STRING(32)	Hyphenated IDs	0.78%	0.77%	+1.0%	0 MC-DQ-10441 A..
referral_code	CRM-Core	STRING(32)	Trailing spaces	7.44%	7.31%	-0.4%	0 MC-DQ-10442 T..
loyalty_tier	LoyaltySvc	ENUM [Bronze,Sil..	Mixed case	0.00%	0.00%	+0.0%	12 MC-DQ-10443 ..
loyalty_points	LoyaltySvc	INTEGER	Negatives	0.21%	0.18%	-1.2%	7 MC-DQ-10444 Fl..
churn_risk_bucket	Analytics	ENUM [Low,Med,..	'Medium' variant	0.00%	0.00%	+0.0%	8 MC-DQ-10445 M..
kyc_status	ComplianceHub	ENUM [Pending,P..	Extra 'Review'	0.11%	0.10%	-0.9%	6 MC-DQ-10446 A..
pep_flag	ComplianceHub	BOOLEAN	0/1 integers	0.00%	0.00%	+0.0%	5 MC-DQ-10447 N..
aml_flag	ComplianceHub	BOOLEAN	0/1 integers	0.00%	0.00%	+0.0%	4 MC-DQ-10448 N..
consent_cookies	OutreachHub	BOOLEAN	Null where not trac..	12.9%	13.2%	+2.1%	0 MC-DQ-10449 A..
device_id	Events-EDL	STRING(64)	Empty strings	3.05%	2.92%	-1.4%	29 MC-DQ-10450 ..
app_version	Events-EDL	STRING(16)	'v' prefix	5.44%	5.31%	-0.7%	0 MC-DQ-10451 S..
web_session_id	Web-EDL	STRING(64)	Mixed entropy	1.20%	1.16%	-0.9%	0 MC-DQ-10452 A..
preferred_language	CRM-Core	STRING ISO-639-1	ISO-639-2 codes	0.83%	0.79%	-0.5%	17 MC-DQ-10453 ..
time_zone	CRM-Core	IANA TZ	GMT offsets	2.34%	2.22%	-1.1%	31 MC-DQ-10454 ..
has_mortgage	Ledger-360	BOOLEAN	'N/A' strings	0.00%	0.00%	+0.0%	12 MC-DQ-10455 ..
has_brokerage	Ledger-360	BOOLEAN	'Unknown' variant	0.00%	0.00%	+0.0%	9 MC-DQ-10456 M..
product_list_count	Analytics	INTEGER	Floats casted	0.14%	0.12%	-1.3%	22 MC-DQ-10457 ..
nps_score	OutreachHub	INTEGER -100..100	Out-of-range 120	0.00%	0.00%	+0.0%	2 MC-DQ-10458 E..
complaint_flag	ServiceDesk	BOOLEAN	'Open'/'Closed'	0.00%	0.00%	+0.0%	13 MC-DQ-10459 I..

support_tickets_op..	ServiceDesk	INTEGER	Negatives	0.03%	0.02%	-1.9%	1 MC-DQ-10460 C..
last_support_intera..	ServiceDesk	TIMESTAMP UTC	Local time	0.49%	0.44%	-1.0%	10 MC-DQ-10461 ..
avg_monthly_spen..	Ledger-360	DECIMAL(18,2)	Rounding diff	0.62%	0.61%	+0.3%	0 MC-DQ-10462 Al..
credit_rating_band	RiskHub	ENUM [A,B,C,D,E]	Lowercase	0.00%	0.00%	+0.0%	16 MC-DQ-10463 ..

Conformance Summary and Ownership

Area	Checks Run	Fields Flagged	SLA for Remediat..	Current On Track	Primary Owner T..	Last Review Time..
Identity & Contact	124	12	10 business days	Yes	Data Engineering (..	2026-06-28T14:52..
Compliance & Risk	76	8	15 business days	Yes	Risk Data Services	2026-06-28T15:11..
Engagement & Lo..	88	10	12 business days	Yes	Customer Analytics	2026-06-28T16:03..
Finance & Value	69	6	10 business days	Yes	Finance Data Ops	2026-06-28T16:27..
Service & Support	41	5	12 business days	Yes	Platform Ops (Ser..	2026-06-28T17:09..

Remediation Tracker (High-Impact Only)

Ticket Ref	Severity	Field(s)	Root Cause Sum..	Target Release Tr..	Status	Owner
MC-DQ-10423	High	email_address	Inconsistent Outre..	CA-Release-2026...	In Progress	OutreachHub Eng
MC-DQ-10431	High	postal_code	Locale-specific for..	CA-Release-2026...	Planned	Data Eng (Ingest)
MC-DQ-10435	High	lifetime_value_usd	Ledger adjustment..	CA-Release-2026...	In Review	Finance Data Ops
MC-DQ-10454	High	time_zone	Legacy CRM expo..	CA-Release-2026...	In Progress	CRM-Core Eng

QA Sign-off: "Findings are consistent with prior quarter and within control tolerances. No blocking defects for v5 schema adoption." -- QA Lead, 2026-06-30T18:42:12Z.

ETL Schema Specification -- Customer Analytics v5

Meridian Capital

Page 16 -- Section 9: Load Window Performance and SLA Compliance (June 2026)

This section summarizes end-to-end pipeline performance for the Customer Analytics v5 paths (extract, conform, enrich, load) during June 2026. Metrics reflect scheduled runs on the daily 02:30 UTC window for Incremental loads and the weekly Sunday 03:00 UTC Full refresh. SLA targets: Daily Incremental <= 22:00 (mm:ss), Weekly Full <= 58:00 (mm:ss). The Data Platform SRE team triangulated durations using orchestrator telemetry and warehouse job analytics. All misses were triaged within four hours per Ops Runbook SLOs. The tables below include load-level metrics, step-wise breakdowns, and watermark alignment versus source systems.

Daily/Weekly Load Performance (June 2026)

Load Date (UTC)	Pipeline Executio..	Batch Type	Records Extracted	Records Loaded	Rejects	Duration (mm:ss)	Avg Throughput (..
2026-06-01	CA-ETL-EXE-2026..	Incremental	5,412,983	5,409,876	3,107	18:44	4,815 22:00 Y - M..
2026-06-02	CA-ETL-EXE-2026..	Incremental	5,506,221	5,503,009	3,212	19:02	4,820 22:00 Y - M..
2026-06-03	CA-ETL-EXE-2026..	Incremental	5,488,577	5,485,212	3,365	19:11	4,775 22:00 Y - M..
2026-06-04	CA-ETL-EXE-2026..	Incremental	5,601,114	5,597,885	3,229	19:24	4,812 22:00 Y - M..
2026-06-05	CA-ETL-EXE-2026..	Incremental	5,593,440	5,590,031	3,409	19:19	4,826 22:00 Y - M..
2026-06-06	CA-ETL-EXE-2026..	Full	78,204,331	78,197,902	6,429	54:37	23,862 58:00 Y - M..
2026-06-07	CA-ETL-EXE-2026..	Incremental	5,481,993	5,478,706	3,287	19:07	4,773 22:00 Y - M..
2026-06-08	CA-ETL-EXE-2026..	Incremental	5,525,610	5,522,287	3,323	19:16	4,777 22:00 Y - M..
2026-06-09	CA-ETL-EXE-2026..	Incremental	5,612,104	5,608,691	3,413	19:28	4,802 22:00 Y - M..
2026-06-10	CA-ETL-EXE-2026..	Incremental	5,644,199	5,640,731	3,468	20:04	4,688 22:00 Y - M..
2026-06-11	CA-ETL-EXE-2026..	Incremental	5,701,008	5,697,545	3,463	20:12	4,700 22:00 Y - M..
2026-06-12	CA-ETL-EXE-2026..	Incremental	5,688,410	5,685,019	3,391	20:09	4,706 22:00 Y - M..
2026-06-13	CA-ETL-EXE-2026..	Full	78,339,112	78,332,608	6,504	55:11	23,668 58:00 Y - M..
2026-06-14	CA-ETL-EXE-2026..	Incremental	5,492,882	5,489,558	3,324	19:23	4,720 22:00 Y - M..
2026-06-15	CA-ETL-EXE-2026..	Incremental	5,541,017	5,537,731	3,286	19:31	4,732 22:00 Y - M..
2026-06-16	CA-ETL-EXE-2026..	Incremental	5,593,772	5,590,384	3,388	19:46	4,716 22:00 Y - M..
2026-06-17	CA-ETL-EXE-2026..	Incremental	5,612,992	5,609,544	3,448	19:59	4,683 22:00 Y - M..
2026-06-18	CA-ETL-EXE-2026..	Incremental	5,655,104	5,651,723	3,381	20:07	4,683 22:00 Y - M..
2026-06-19	CA-ETL-EXE-2026..	Incremental	5,671,009	5,667,518	3,491	20:16	4,664 22:00 Y - M..
2026-06-20	CA-ETL-EXE-2026..	Full	78,471,883	78,465,102	6,781	56:09	23,287 58:00 Y - M..
2026-06-21	CA-ETL-EXE-2026..	Incremental	5,489,901	5,486,541	3,360	19:28	4,699 22:00 Y - M..
2026-06-22	CA-ETL-EXE-2026..	Incremental	5,538,772	5,535,449	3,323	19:35	4,713 22:00 Y - M..
2026-06-23	CA-ETL-EXE-2026..	Incremental	5,599,447	5,596,177	3,270	19:53	4,687 22:00 Y - M..
2026-06-24	CA-ETL-EXE-2026..	Incremental	5,641,100	5,637,718	3,382	20:06	4,676 22:00 Y - M..
2026-06-25	CA-ETL-EXE-2026..	Incremental	5,682,771	5,679,299	3,472	20:18	4,662 22:00 Y - M..
2026-06-26	CA-ETL-EXE-2026..	Incremental	5,711,388	5,707,957	3,431	20:21	4,671 22:00 Y - M..
2026-06-27	CA-ETL-EXE-2026..	Full	78,609,550	78,602,731	6,819	56:24	23,237 58:00 Y - M..
2026-06-28	CA-ETL-EXE-2026..	Incremental	5,498,882	5,495,540	3,342	19:36	4,672 22:00 Y - M..
2026-06-29	CA-ETL-EXE-2026..	Incremental	5,557,110	5,553,801	3,309	19:42	4,700 22:00 Y - M..
2026-06-30	CA-ETL-EXE-2026..	Incremental	5,611,728	5,608,361	3,367	20:03	4,663 22:00 Y - M..

Step-Level Duration Breakdown (June Mean; v5 Pipelines)

Step Order	Transformation S..	Step Identifier	Mean Duration (s)	P95 Duration (s)	Rows Processed ..	Error Rows (Mean)	Notes Owner Team
1	Source Extract -- C..	STEP-EX-CRM	148	179	2,104,300	1,011	Stable; minor sche..
2	Source Extract -- L..	STEP-EX-LED	201	238	1,987,500	903	Consistent payload..
3	Source Extract -- ..	STEP-EX-OUT	156	191	1,321,100	1,214	Email validator upd..
4	Conform -- Type C..	STEP-CF-TYPE	226	271	5,412,900	3,310	Improved null nor..

5	Enrich -- Referenc..	STEP-EN-ISO	119	145	5,412,900	61	New ISO table v20..
6	Enrich -- Risk Scor..	STEP-EN-RISK	133	164	5,412,900	17	RiskHub feed stabl..
7	Enrich -- Loyalty	STEP-EN-LTY	94	121	5,412,900	9	Title-case normaliz..
8	Deduplicate -- Cus..	STEP-DD-CUST	102	128	5,409,800	0	Deterministic hash ..
9	Load -- Warehous..	STEP-LD-STG	167	199	5,409,800	0	Staging compactio..
10	Load -- Warehous..	STEP-LD-FDW	213	259	5,409,800	0	Partition pruning ef..
11	Post-Load -- Qualit..	STEP-QL-GATE	74	97	5,409,800	3,214	Gate failures trend ..

Incremental Watermark Alignment (as of 2026-06-30T06:00:00Z)

Source Entity	Watermark Colu..	Last Watermark V..	Lag vs Source	Expected Lag SLA	Status	Ticket
CRM-Core.custom..	updated_at (UTC)	2026-06-30T02:17..	13m	<= 20m	On Target	MC-OPS-32748
Ledger-360.transa..	posted_ts (UTC)	2026-06-30T02:08..	22m	<= 30m	On Target	MC-OPS-32749
OutreachHub.cont..	last_modified (UTC)	2026-06-30T02:12..	18m	<= 25m	On Target	MC-OPS-32750
Events-EDL.sessio..	event_ts (UTC)	2026-06-30T02:05..	26m	<= 30m	On Target	MC-OPS-32751
RiskHub.scores	score_ts (UTC)	2026-06-30T01:59..	31m	<= 35m	On Target	MC-OPS-32752
ServiceDesk.cases	modified_ts (UTC)	2026-06-30T02:14..	16m	<= 25m	On Target	MC-OPS-32753

Observations and Actions

- All 30 June daily incrementals met SLA with mean duration 19:47 (mm:ss); weekly full refreshes averaged 55:58 with no anomalies.
- Reject rates averaged 0.058% of extracted rows, dominated by email normalization and postal code canonicalization (tracked via MC-DQ-10423 and MC-DQ-10431).
- Step-level p95s show headroom against targets; no single step exceeded 5 minutes at p95 within incrementals.
- Watermarks remained within defined lags; longest observed was RiskHub at 31m, within SLA.
- Action items: finalize OutreachHub email validator rollout (CA-Release-2026.28), complete postal code canonicalizer improvements (CA-Release-2026.29), and maintain weekly review cadence with Platform Ops for staging compaction tuning.

SRE Compliance Note: "June run profile demonstrates stable throughput and predictable variance. No capacity changes recommended before Q3 peak week." -- Platform SRE Duty Manager, 2026-07-01T07:21:33Z.

Page 17 -- Data Quality Validation and Profiling Results (Sprint 49, Customer Analytics v5)

The Customer Analytics v5 data quality review for Sprint 49 was executed by the Data Quality Office in collaboration with Customer Analytics QA and Data Platform Operations. Validations were performed in UAT-NA and Prod-EU partitions between 2026-07-11 and 2026-07-19, using the standardized rules catalog v5.12. Thresholds reflect CAB-agreed tolerance bands documented in MC-ANL-4701. The tables below summarize rule outcomes and field-level profiling highlights for priority entities feeding KPI dashboards and analytic models. No gating defects were observed on core dimensions; two non-gating items remain open pending routine backlog remediation. Evidence datasets and sample IDs are retained in S3 analytics-audit buckets under the v5 suffix identifiers with retention labels per records policy.

DQ Check Summary -- Sprint 49

Check_ID	Dataset	Dimension	Rule_Description	Threshold	Observed	Status	Sample_Ref Run_..
DQ-049-001	dim_customer_v5	Completeness	customer_sk not n..	100%	100%	Pass	SA-11293 2026-07..
DQ-049-002	dim_customer_v5	Uniqueness	customer_sk unique	100%	100%	Pass	SA-11294 2026-07..
DQ-049-003	dim_customer_v5	Conformity	country_code in IS..	>= 99.9%	99.98%	Pass	SA-11295 2026-07..
DQ-049-004	dim_customer_v5	Validity	birth_date <= load..	100%	100%	Pass	SA-11296 2026-07..
DQ-049-005	fact_interaction_v5	Referential Integrity	customer_sk exists..	100%	100%	Pass	SA-11297 2026-07..
DQ-049-006	fact_interaction_v5	Timeliness	event_ts within 2h ..	>= 99.5%	99.7%	Pass	SA-11298 2026-07..
DQ-049-007	fact_interaction_v5	Conformity	channel in {web, m..	>= 99.9%	99.92%	Pass	SA-11299 2026-07..
DQ-049-008	brdg_customer_se..	Completeness	segment_sk not null	100%	100%	Pass	SA-11301 2026-07..
DQ-049-009	brdg_customer_se..	Referential Integrity	segment_sk exists ..	100%	100%	Pass	SA-11302 2026-07..
DQ-049-010	dim_product_v5	Conformity	product_status in {..	>= 99.9%	100%	Pass	SA-11303 2026-07..
DQ-049-011	dim_product_v5	Uniqueness	product_code uniq..	100%	100%	Pass	SA-11304 2026-07..
DQ-049-012	fact_channel_attrib..	Completeness	attribution_weight ..	100%	100%	Pass	SA-11306 2026-07..
DQ-049-013	fact_channel_attrib..	Validity	0 <= attribution_we..	100%	100%	Pass	SA-11307 2026-07..
DQ-049-014	xref_customer_gui..	Conformity	guid format match..	>= 99.9%	99.99%	Pass	SA-11308 2026-07..
DQ-049-015	stg_customer_raw..	Completeness	email_domain not ..	>= 99.5%	99.4%	Warn	SA-11310 2026-07..
DQ-049-016	stg_customer_raw..	Conformity	phone_country_co..	>= 99.5%	99.8%	Pass	SA-11311 2026-07..
DQ-049-017	dim_customer_v5	Timeliness	daily snapshot rea..	100%	100%	Pass	SA-11312 2026-07..
DQ-049-018	fact_interaction_v5	Volume	expected +/- 3% vs..	within band	within band	Pass	SA-11313 2026-07..
DQ-049-019	dim_segment_v5	Completeness	segment_name no..	100%	100%	Pass	SA-11314 2026-07..
DQ-049-020	dim_customer_v5	Conformity	loyalty_tier in v5 e..	>= 99.9%	99.98%	Pass	SA-11315 2026-07..
DQ-049-021	fact_product_enga..	Validity	session_duration_..	100%	100%	Pass	SA-11316 2026-07..
DQ-049-022	fact_product_enga..	Conformity	device_type in {iO..	>= 99.5%	99.6%	Pass	SA-11317 2026-07..
DQ-049-023	dim_customer_v5	Reasonableness	age between 18 an..	>= 99.8%	99.85%	Pass	SA-11318 2026-07..
DQ-049-024	fact_channel_attrib..	Consistency	sum(weights) per c..	>= 99.9%	99.92%	Pass	SA-11319 2026-07..

Field-Level Profiling -- Selected Columns (Profile Window: 2026-07-10 to 2026-07-16)

Dataset	Column	Data_Type	Distinct_Count	Null_Rate	Min	Max	Mean StdDev Top..
dim_customer_v5	customer_sk	BIGINT	12,498,221	0.00%	1	17,913,552	-- -- -- surrogate ke..
dim_customer_v5	country_code	CHAR(2)	62	0.01%	--	--	-- -- US, GB, DE, C..
dim_customer_v5	loyalty_tier	VARCHAR(12)	5	0.02%	--	--	-- -- Silver, Gold, Pl..
dim_customer_v5	birth_date	DATE	10,921	0.47%	1916-01-01	2008-12-31	-- -- -- nulls within t..
fact_interaction_v5	interaction_id	BIGINT	178,221,093	0.00%	1001	412,882,991	-- -- -- sequence m..
fact_interaction_v5	event_ts	TIMESTAMP	6,912,204	0.00%	2026-06-10T00:00:..	2026-07-16T23:59:..	-- -- -- time span ali..
fact_interaction_v5	channel	VARCHAR(8)	4	0.00%	--	--	-- -- web, mobile, b..
fact_interaction_v5	session_duration_..	INT	18,112	0.12%	0	7,214	322.4 119.7 -- long..
brdg_customer_se..	segment_sk	INT	48	0.00%	1	64	-- -- -- stable acros..
brdg_customer_se..	active_flag	BOOLEAN	2	0.00%	--	--	-- -- true, false true..
dim_product_v5	product_code	VARCHAR(24)	1,202	0.00%	--	--	-- -- PCHK, PBRK, ..
dim_product_v5	product_status	VARCHAR(8)	3	0.00%	--	--	-- -- active, inactive..
xref_customer_gui..	guid	UUID	12,498,219	0.00%	--	--	-- -- -- 2 formatting ..

fact_channel_attrib..	attribution_weight	DECIMAL(5,4)	1,001	0.00%	0.0000	1.0000	0.2500 0.1442 -- ro..
fact_channel_attrib..	model_version	VARCHAR(10)	3	0.00%	--	--	-- -- v5.1, v5.2, v5...
stg_customer_raw..	email_domain	VARCHAR(128)	8,144	0.61%	--	--	-- -- example.com, ..
stg_customer_raw..	phone_country_co..	VARCHAR(4)	41	0.07%	1	998	-- -- -- conformity i..
fact_product_enga..	device_type	VARCHAR(8)	4	0.00%	--	--	-- -- iOS, Android, ..
fact_product_enga..	clicks	INT	1,712	0.00%	0	2,487	3.22 5.91 -- zero-in..
dim_segment_v5	segment_name	VARCHAR(64)	48	0.00%	--	--	-- -- Mass, Affluent,..

Open Quality Findings -- Snapshot (Non-Gating)

Defect_ID	Severity	Brief_Description	Dataset	Impacted_Rows	Workaround	Target_Fix_Sprint	Assignee Status ..
DQDEF-254	Low	email_domain null..	stg_customer_raw..	15,128	downstream treats ..	S50	CA-QA (S. Ibarra) I..
DQDEF-261	Low	rare device_type la..	fact_product_enga..	2,913	remap view applie..	S50	DQ Office (L. Men..

Page 18 -- Schema Change Review and Downstream Impact (H2 FY26, Customer Analytics v5)

This section records the schema change proposals assessed by the Change Advisory Board (CAB) for Customer Analytics v5 during H2 FY26 planning, along with downstream consumer impacts and operational throughput metrics. Changes align with model evolution tickets and stewardship requests. All approved items include effective dates and owners; proposals remain in "Under Review" until validation evidence meets acceptance criteria in UAT. Communications were issued through the Enterprise Release Calendar and the Analytics Change Bulletin. No consumer experienced material SLA degradation during the reviewed window.

Schema Change Review Log (v5 Scope)

Change_ID	Object	Change_Type	Description	Status	Target_Release	Risk	Owner_Team CA..
SCH-5901	dim_customer_v5	Add Column	add marketing_opt..	Approved	5.2.2	Low	Customer Analytic..
SCH-5902	dim_customer_v5	Modify Column	widen loyalty_tier f..	Approved	5.2.2	Low	Customer Analytic..
SCH-5907	fact_interaction_v5	Add Column	add referrer_host ..	Approved	5.2.3	Medium	Data Platform Ops ..
SCH-5910	fact_interaction_v5	Deprecate Column	deprecate legacy s..	Approved	5.3.0	Medium	Data Platform Ops ..
SCH-5914	dim_product_v5	Add Column	add regulatory_cat..	Approved	5.2.3	Low	Customer Analytic..
SCH-5916	brdg_customer_se..	Modify PK	change PK from (c..	Under Review	5.3.0	High	Customer Analytic..
SCH-5919	fact_channel_attrib..	Add Column	add model_confide..	Approved	5.2.4	Medium	Customer Analytic..
SCH-5921	xref_customer_gui..	Add Constraint	enforce UUIDv4 ch..	Approved	5.2.2	Low	Data Quality Office..
SCH-5924	dim_segment_v5	Add Column	add segment_prior..	Under Review	5.2.4	Low	Customer Analytic..
SCH-5927	fact_product_enga..	Modify Column	increase clicks INT..	Approved	5.3.0	Medium	Data Platform Ops ..
SCH-5931	dim_customer_v5	Add Index	nonclustered on (c..	Approved	5.2.3	Low	Data Platform Ops ..
SCH-5933	fact_interaction_v5	Rename Column	rename ua_device ..	Under Review	5.3.1	Medium	Customer Analytic..
SCH-5936	dim_product_v5	Deprecate Column	deprecate legacy p..	Approved	5.3.0	Low	Customer Analytic..
SCH-5940	fact_channel_attrib..	Modify Column	widen attribution_..	Under Review	5.3.1	Low	Customer Analytic..
SCH-5943	stg_customer_raw..	Add Column	add source_partne..	Approved	5.2.2	Low	Data Platform Ops ..
SCH-5947	dim_customer_v5	Add SCD Attribute	add preferred_lang..	Under Review	5.3.1	Medium	Customer Analytic..
SCH-5950	brdg_customer_se..	Add Column	add membership_s..	Approved	5.2.4	Medium	Customer Analytic..
SCH-5953	dim_segment_v5	Modify Column	widen segment_na..	Approved	5.2.4	Low	Customer Analytic..

Downstream Consumer Impact Matrix (Consumer Readiness and Communications)

Consumer_System	Owner	Interface_Type	Fields_Impacted	Action_Required	SLA_Risk	Notice_Sent	Ticket_Ref Comm..
Marketing Hub	Growth Tech	Snowflake View	marketing_opt_in_dt	schema alignment ..	None	2026-07-19	MC-ANL-4861 non..
Personalization API	Digital Platform	API via View	user_agent_device..	monitor only	Low	2026-07-21	MC-ANL-4870 alia..
Finance MI	Finance Analytics	Extract (CSV)	segment_priority (..	version bump	None	2026-07-20	MC-ANL-4865 will ..
Risk Scoring	Risk Tech	Snowflake Table	model_confidence	no change	None	2026-07-19	MC-ANL-4862 field..
Contact Center BI	Client Services	PowerBI DirectQu..	referrer_host	schema alignment ..	Low	2026-07-22	MC-ANL-4873 das..
Campaign Orchest..	Growth Ops	S3 Parquet	membership_stren..	monitor only	Low	2026-07-19	MC-ANL-4863 dow..
Partner Reporting	Partnerships	SFTP CSV	source_partner_id	version bump	Low	2026-07-18	MC-ANL-4859 pas..
Mobile App Telem..	Mobile Eng	Snowflake View	device_type remap	no change	None	2026-07-20	MC-ANL-4866 ma..
Web Analytics	Digital Analytics	Snowflake View	referrer_host	schema alignment ..	Low	2026-07-22	MC-ANL-4872 join ..
Data Science Lab	Advanced Analytics	Notebook (SQL)	attribution_weight ..	monitor only	None	2026-07-21	MC-ANL-4869 no r..
Investor Relations	Corporate	PDF Report	segment_name len..	no change	None	2026-07-19	MC-ANL-4864 title..
CRM Sync	Enterprise Integrati..	API via View	loyalty_tier width	no change	None	2026-07-18	MC-ANL-4858 bin..
Sales Insights	Revenue Ops	Tableau Live	model_confidence	monitor only	Low	2026-07-20	MC-ANL-4867 visu..
Client Onboarding	Onboarding Tech	Snowflake Table	preferred_language	schema alignment ..	Low	2026-07-23	MC-ANL-4876 pen..
Attribution Service	Growth Science	API via View	attribution_weight ..	no change	None	2026-07-21	MC-ANL-4871 JS..
Enterprise Lake Ex..	Data Platform	S3 Parquet	session_id_txt (de..	monitor only	Low	2026-07-22	MC-ANL-4874 mar..

Operational Throughput Metrics -- Week 29 (Mon 2026-07-13 to Sun 2026-07-19)

Job_ID	Pipeline	Avg_Duration_mm	P95_mm	Rows_Processed..	Success_Rate	Last_Failure	Incident_Ref Own..
ETL-CUST-101	dim_customer_v5_..	42	58	12.6	100.0%	--	-- ETL Operations
ETL-INTR-205	fact_interaction_v5..	73	98	178.2	99.9%	2026-07-14T05:12Z	INC-CA-731 Data ..
ETL-SEG-310	brdg_customer_se..	27	35	24.4	100.0%	--	-- Customer Analyt..
ETL-PROD-121	dim_product_v5_s..	18	24	1.2	100.0%	--	-- Customer Analyt..
ETL-ATTR-415	fact_channel_attrib..	66	82	9.8	100.0%	--	-- Growth Science ..
ETL-GUID-509	xref_customer_gui..	12	19	12.5	100.0%	--	-- Data Quality Offi..
ETL-RAW-603	stg_customer_raw..	34	48	13.9	99.8%	2026-07-16T02:01Z	INC-CA-734 Data ..
ETL-ENG-702	fact_product_enga..	39	55	44.1	100.0%	--	-- ETL Operations
ETL-SEG-809	dim_segment_v5_..	9	12	0.04	100.0%	--	-- Customer Analyt..
ETL-ANL-910	analytics_mart_roll..	58	73	221.7	100.0%	--	-- Customer Analyt..
ETL-CUST-111	dim_customer_v5_..	7	9	n/a	100.0%	--	-- Data Platform O..
ETL-ATTR-419	attribution_export_..	21	29	9.8	100.0%	--	-- Growth Ops Tech

Note: Two transient failures (INC-CA-731, INC-CA-734) were auto-recovered within the window without breaching consumer SLAs. CAB approvals for 5.2.2 are synchronized to the Enterprise Release Calendar on 2026-08-05, with consumer notices already issued under the referenced tickets.

Page 19 -- Data Quality Thresholds, Reviews, and KPI Trends

This page consolidates the v5 data quality thresholds adopted for Customer Analytics entities, with supporting review logs and KPI trend snapshots. Thresholds reflect a balance between analytical utility and source system variability, and are monitored via scheduled data quality (DQ) jobs in the shared platform. Findings documented here were remediated within normal SLA windows and are retained for auditability. All timestamps are UTC and reference the production lineage for the ETL Schema Specification -- Customer Analytics v5 (release train CA-2026.07).

Data Quality Threshold Register (v5 Effective Window 2026-06-01 to 2027-05-31)

Dataset	Column	Check Type	Threshold	Sampling Scope	Monitoring Job ID	Owner Team	Ticket Ref (Last U..
ca_customer	customer_id	Uniqueness	100.00% unique	Full dataset	dq_ca_cust_01	DE-CA	MC-ANL-15432 20..
ca_customer	email_hash	Completeness	>= 99.50% non-null	Incremental loads	dq_ca_cust_02	DE-CA	MC-ANL-15601 20..
ca_customer	birth_date	Valid Range	1900-01-01 to now()	Full dataset	dq_ca_cust_03	DGO	DQ-CA-442 2026-..
ca_customer	country_code	Domain	ISO 3166-1 alpha-2	Incremental loads	dq_ca_cust_04	DGO	MC-DATA-7712 20..
ca_account	account_id	Uniqueness	100.00% unique	Full dataset	dq_ca_acct_01	DE-CA	MC-ANL-15512 20..
ca_account	customer_id	Referential Integrity	100.00% to ca_cus..	Full dataset	dq_ca_acct_02	DE-CA	MC-ETL-43210 20..
ca_account	open_date	Completeness	>= 99.90% non-null	Incremental loads	dq_ca_acct_03	DE-CA	MC-ANL-15644 20..
ca_transaction	txn_id	Uniqueness	100.00% unique	Full dataset	dq_ca_txn_01	DE-CA	MC-ANL-15678 20..
ca_transaction	txn_ts	Timeliness	<= 5 min ingestion ..	Streaming micro-b..	dq_ca_txn_02	SRE-DP	OPS-DP-9821 202..
ca_transaction	amount	Format/Type	Numeric, 2 decimals	Full dataset	dq_ca_txn_03	DGO	DQ-CA-455 2026-..
ca_transaction	currency	Domain	ISO 4217	Incremental loads	dq_ca_txn_04	DGO	MC-DATA-7729 20..
ca_segmentation	segment_code	Domain	Catalog v5.1	Full dataset	dq_ca_seg_01	Customer Insights	CI-REF-223 2026-..
ca_segmentation	effective_from	Completeness	100.00% non-null	Full dataset	dq_ca_seg_02	DE-CA	MC-ANL-15681 20..
ca_marketing_optin	opt_in_flag	Valid Set	{Y, N}	Incremental loads	dq_ca_mkt_01	DGO	DQ-CA-463 2026-..
ca_marketing_optin	source_system	Domain	Catalog v5 Systems	Incremental loads	dq_ca_mkt_02	DGO	MC-DATA-7751 20..
ca_ltv	ltv_12m	Statistical Bounds	Z-score <= 4.0	Full dataset	dq_ca_ltv_01	DE-CA	MC-ANL-15710 20..
ca_ltv	model_version	Completeness	100.00% non-null	Incremental loads	dq_ca_ltv_02	Customer Insights	CI-ML-581 2026-0..
ca_attribution	touchpoint_ts	Sequence	Non-decreasing pe..	Incremental loads	dq_ca_attr_01	DE-CA	MC-ANL-15722 20..
ca_attribution	channel	Domain	Marketing Channel..	Full dataset	dq_ca_attr_02	DGO	DQ-CA-471 2026-..
ca_consent	consent_ts	Completeness	>= 99.80% non-null	Incremental loads	dq_ca_con_01	DGO	MC-DATA-7762 20..

Data Quality Review Log (Post Go-Live Observation Window: 2026-06-01 to 2026-07-15)

Review Window (..	Check	Finding ID	Severity	Affected Rows %	Root Cause Sum..	Remediation Tick..	Disposition Verifi..
2026-06-03T12:00:..	ca_customer.email..	DQF-CA-001	Low	0.38%	Source feed interm..	MC-ANL-15601	Closed DGO 2026-..
2026-06-05T09:30:..	ca_account.custo..	DQF-CA-002	Medium	0.06%	Late-arriving custo..	MC-ETL-43210	Closed DE-CA 202..
2026-06-06T21:45:..	ca_transaction.txn..	DQF-CA-003	Low	n/a	Streaming micro-b..	OPS-DP-9821	Closed SRE-DP 2..
2026-06-08T14:10:..	ca_customer.count..	DQF-CA-004	Low	0.02%	Non-catalog test c..	MC-DATA-7712	Closed DGO 2026-..
2026-06-10T07:50:..	ca_segmentation.s..	DQF-CA-005	Low	0.11%	Segment catalog v..	CI-REF-223	Closed Customer I..
2026-06-11T12:20:..	ca_account.open_..	DQF-CA-006	Low	0.07%	Stale branch batch..	MC-ANL-15644	Closed DE-CA 202..
2026-06-12T23:10:..	ca_transaction.curr..	DQF-CA-007	Low	0.01%	Obsolete currency ..	MC-DATA-7729	Closed DGO 2026-..
2026-06-14T05:00:..	ca_marketing_opti..	DQF-CA-008	Low	0.05%	Upstream boolean ..	DQ-CA-463	Closed DE-CA 202..
2026-06-16T18:40:..	ca_ltv.ltv_12m bou..	DQF-CA-009	Medium	0.13%	Promotional cohort..	MC-ANL-15710	Closed Customer I..
2026-06-18T11:05:..	ca_attribution.touc..	DQF-CA-010	Low	0.09%	Event timestamp p..	MC-ANL-15722	Closed DE-CA 202..
2026-06-19T16:25:..	ca_consent.conse..	DQF-CA-011	Low	0.12%	Minor nulls due to ..	MC-DATA-7762	Closed DGO 2026-..
2026-06-22T13:15:..	ca_transaction.am..	DQF-CA-012	Low	0.03%	Locale decimal deli..	DQ-CA-455	Closed DE-CA 202..
2026-06-25T08:00:..	ca_customer.birth_..	DQF-CA-013	Low	0.01%	Default placeholde..	DQ-CA-442	Closed DGO 2026-..
2026-06-28T19:30:..	ca_segmentation.e..	DQF-CA-014	Low	0.00%	No issue; quarterly..	MC-ANL-15681	Closed DE-CA 202..
2026-07-01T10:05:..	ca_transaction.txn..	DQF-CA-015	Low	n/a	Short lag spike cor..	OPS-DP-9821	Closed SRE-DP 2..

KPI Trend Snapshot (Monthly Rolling, Q2-Q3 FY2026)

KPI	Oct	Nov	Dec	Jan	Feb	Mar	Trend Notes Data..
Customer Key Uni..	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	Stable No duplicat..

Email Hash Compl..	99.46%	99.51%	99.53%	99.54%	99.55%	99.56%	Improving Vendor ..
Account-Customer..	99.94%	99.97%	100.00%	100.00%	100.00%	100.00%	Improving Late-arri..
Txn Timeliness p9..	4.2	4.1	4.0	3.9	4.0	3.8	Stable Within SLA ..
Currency Domain ..	99.99%	100.00%	100.00%	100.00%	100.00%	100.00%	Stable Legacy cod..
Segment Code Co..	99.80%	99.88%	99.92%	99.95%	99.96%	99.97%	Improving Cache r..
Opt-in Flag Validity	99.94%	99.95%	99.96%	99.97%	99.98%	99.98%	Improving Case no..
LTV Outlier Rate	0.21%	0.19%	0.17%	0.16%	0.15%	0.14%	Improving Z-score ..
Attribution Sequen..	99.70%	99.78%	99.84%	99.90%	99.92%	99.93%	Improving Timesta..
Consent Timestam..	99.72%	99.78%	99.81%	99.83%	99.85%	99.86%	Improving Parser h..

Page 20 -- Change Log, Operational Readiness, and Run Metrics

This page finalizes the v5 specification with a documented change log of schema elements, operational readiness controls, and a rollup of early run metrics following go-live. Each change record links to its authoritative ticket, with explicit indicators of backward compatibility and report impact. Readiness controls reflect standard enterprise gates applied by Data Engineering, Data Governance, and SRE for production acceptance. The metrics rollup provides a compact view of throughput and SLA adherence for the initial stabilization period.

Schema Change Log -- Customer Analytics v5 (Release Train CA-2026.07)

Change ID	Object Type	Object Name	Change Descripti..	Backward Compa..	Related Tickets	Requested By	Impacted Reports..
CHG-CA-5001	Table	ca_customer	Added email_hash..	Backward compati..	MC-ANL-15432	Customer Insights	MKT-Engagement ..
CHG-CA-5002	Column	ca_customer.count..	Constrained domai..	Requires alias rem..	MC-DATA-7712	Data Governance	Intl Segmentation ..
CHG-CA-5003	Table	ca_account	Enforced RI to ca_..	Backward compati..	MC-ETL-43210	DE-CA	Account Summary ..
CHG-CA-5004	Column	ca_account.open_..	Standardized to D..	Backward compati..	MC-ANL-15644	DE-CA	Account Aging v2 ..
CHG-CA-5005	Table	ca_transaction	Introduced streami..	Backward compati..	OPS-DP-9821	SRE-DP	Transaction Feed ..
CHG-CA-5006	Column	ca_transaction.curr..	Aligned to ISO 421..	Backward compati..	MC-DATA-7729	DGO	FX Exposure v1 v5..
CHG-CA-5007	Column	ca_transaction.am..	Cast to DECIMAL(..	Backward compati..	DQ-CA-455	DE-CA	Revenue Cube v6 ..
CHG-CA-5008	Table	ca_segmentation	Refreshed segmen..	Backward compati..	CI-REF-223	Customer Insights	Segmentation Hub..
CHG-CA-5009	Column	ca_segmentation.e..	Required non-null f..	Backward compati..	MC-ANL-15681	DE-CA	Segmentation Tren..
CHG-CA-5010	Table	ca_marketing_optin	Standardized opt_i..	Backward compati..	DQ-CA-463	DGO	Consent Dashboar..
CHG-CA-5011	Table	ca_ltv	Introduced ltv_12m..	Backward compati..	MC-ANL-15710	Customer Insights	LTV Planner v1 v5...
CHG-CA-5012	Column	ca_ltv.model_versi..	Required attribute f..	Backward compati..	CI-ML-581	Customer Insights	Model Registry v2 ..
CHG-CA-5013	Table	ca_attribution	Enforced non-decr..	Backward compati..	MC-ANL-15722	DE-CA	Attribution Paths v..
CHG-CA-5014	Column	ca_attribution.chan..	Conformed channe..	Backward compati..	DQ-CA-471	DGO	Channel Mix v3 v5...
CHG-CA-5015	Table	ca_consent	Harmonized timest..	Backward compati..	MC-DATA-7762	DGO	Consent Pipeline v..
CHG-CA-5016	Column	ca_customer.birth_..	Applied range valid..	Backward compati..	DQ-CA-442	DGO	Demographic Over..
CHG-CA-5017	Column	ca_customer.email..	Set as alternate bu..	Backward compati..	MC-ANL-15601	DE-CA	Journey 360 v2 v5...
CHG-CA-5018	Ingest	ca_transaction stre..	Increased event-ti..	Backward compati..	OPS-DP-9821	SRE-DP	Real-time KPIs v1 ..

Operational Readiness Checklist and Sign-offs (Production Acceptance CA-2026.07)

Control ID	Control Descripti..	Evidence Referen..	Owner Team	Reviewer	Review Date (UTC)	Status	Notes Ticket
ORC-CA-001	Data dictionary up..	EVD-CA-1001	DGO	Data Governance ..	2026-06-01T11:00:..	Approved	All fields mapped ..
ORC-CA-002	Schema contracts ..	EVD-CA-1002	DE-CA	Customer Analytic..	2026-06-02T15:30:..	Approved	No breaking chang..
ORC-CA-003	Non-functional SL..	EVD-CA-1003	SRE-DP	SRE Manager	2026-06-02T18:15:..	Approved	Throughput and ti..
ORC-CA-004	Monitoring dashbo..	EVD-CA-1004	SRE-DP	Platform Observab..	2026-06-03T09:45:..	Approved	DQ, latency, and v..
ORC-CA-005	Backfill verification ..	EVD-CA-1005	DE-CA	Analytics QA	2026-06-04T14:20:..	Approved	Sample-to-total rec..
ORC-CA-006	Data retention alig..	EVD-CA-1006	DGO	Data Steward	2026-06-05T10:05:..	Approved	Retention windows..
ORC-CA-007	Access entitlement..	EVD-CA-1007	DGO	Access Governance	2026-06-05T16:35:..	Approved	Consumer roles m..
ORC-CA-008	Incident runbooks l..	EVD-CA-1008	SRE-DP	SRE On-call	2026-06-06T08:50:..	Approved	References consoli..
ORC-CA-009	Dependency catal..	EVD-CA-1009	DE-CA	Architecture Board	2026-06-06T13:10:..	Approved	Upstream/downstr..
ORC-CA-010	UAT sign-off from ..	EVD-CA-1010	Customer Insights	Product Owners C..	2026-06-07T12:40:..	Approved	Test cases passed..
ORC-CA-011	Data quality baseli..	EVD-CA-1011	DGO	Data Governance ..	2026-06-08T09:25:..	Approved	Threshold register ..
ORC-CA-012	Release artifact int..	EVD-CA-1012	DE-CA	Release Manager	2026-06-08T17:55:..	Approved	Build manifests co..

Run Metrics Rollup -- Weeks 1-8 Post Go-Live (Production)

Week Ending (UT..	Scheduled Runs	Successful Runs	SLA Met %	Mean Runtime (m..	95th pct (min)	Rows Processed ..	Incidents Primary..
2026-06-07	84	83	98.8%	36	49	1,920	1 OPS-DP-9821 Cl..
2026-06-14	84	84	100.0%	35	47	1,945	0 n/a n/a Stable w..
2026-06-21	84	84	100.0%	35	46	1,951	0 n/a n/a All pipelin..
2026-06-28	84	84	100.0%	34	45	1,958	0 n/a n/a Latency t..

2026-07-05	84	84	100.0%	34	44	1,962	0 n/a n/a Sustaine..
2026-07-12	84	83	98.8%	35	48	1,967	1 OPS-DP-9842 In..
2026-07-19	84	84	100.0%	34	45	1,973	0 n/a n/a Follow-up..
2026-07-26	84	84	100.0%	33	44	1,981	0 n/a n/a Best wee..

End of document -- ETL Schema Specification -- Customer Analytics v5, Meridian Capital.